ENGINEERING GEOLOGY

UNIT - I : INTRODUCTION

Branches, Importance, Weathering, Geological Agents & Rivers

Syllabus (as given)

- Branches of Geology, Importance of Geology in Civil Engineering with case studies, weathering of rocks, Geological agents, weathering process of Rock, Rivers and geological work of rivers.

1. Introduction to Geology

Geology (Greek: geo = earth, logos = discourse) is the science that deals with the origin, composition, structure, history and processes of the Earth.

Engineering Geology = application of geological knowledge to civil engineering works (dams, tunnels, bridges, roads, buildings, reservoirs, canals, slopes).

- Earth facts: age ≈ 4.6 billion years; radius ≈ 6371 km; layers = **crust, mantle, core**
  - Crust: 5-10 km (oceanic), 30-70 km (continental) - mostly granite (SIAL) and basalt (SIMA)
  - Mantle: ≈ 2900 km thick; Core: outer (liquid) + inner (solid, Fe-Ni)
- Geology is an **observational + interpretative** science - rocks are studied in the field, in the lab and by instruments.

1.1 Branches of Geology

1. **Physical Geology** - study of the earth's materials, structure and the processes acting on and within it (weathering, erosion, volcanism, earthquakes).
2. **Mineralogy** - study of minerals: composition, crystal form, properties, origin, uses.
3. **Petrology** - study of rocks: their origin, texture, classification (igneous, sedimentary, metamorphic).
4. **Structural Geology** - deformation of rocks: folds, faults, joints, unconformities.
5. **Stratigraphy** - study of layered rocks, their sequence, age and correlation.
6. **Palaeontology** - study of fossils (ancient life) - used for dating and correlation of

strata.

7. **Historical Geology** - history of the earth and its life through geological time.
8. **Geomorphology** - origin and evolution of landforms (hills, valleys, plains, deltas).
9. **Economic Geology** - ores, coal, petroleum, building stones and other useful materials.
10. **Hydrogeology** - occurrence, movement and quality of ground water.
11. **Engineering Geology** - geology applied to engineering projects and site safety.
12. **Geophysics, Geochemistry, Seismology, Volcanology, Oceanography** - supporting branches.

Quick memory trick

- Minerals -> Rocks -> Structures -> Strata -> Fossils -> Landforms -> Water -> Engineering. (Material first, then arrangement, then time, then use!)

1.2 Importance of Geology in Civil Engineering

Every civil structure rests **on or in** the earth, so its safety depends on the nature of the ground. Geology helps the engineer in:

- **Site selection & investigation** - for dams, reservoirs, tunnels, bridges, highways, airports, power plants.
- **Foundation conditions** - bearing capacity, weak zones, cavities, soft clay, expansive (black cotton) soil, settlement.
- **Construction material** - identifying good sources of stone, sand, gravel, clay, lime, laterite for aggregate and bricks.
- **Water problems** - ground-water level, seepage, leakage from reservoirs, water supply and drainage.
- **Natural hazards** - earthquakes, landslides, floods, subsidence, tsunami: zoning and mitigation.
- **Structures in rocks** - attitude of beds, joints, faults decide stability of slopes, tunnels and dam abutments.
- **Cost & time control** - early geological knowledge avoids redesign, failures and extra expenditure.
- **Mineral & resource exploration** - roads/mines/quarries planning, geophysical surveys.

Remember

- Geologist finds out WHAT the ground is, the engineer decides HOW to build on it. "Geology is the foundation of foundations."

1.3 Case Studies (failures due to geological neglect)

(a) Vajont Dam, Italy (1963)

- 262 m high **double-curvature arch dam** - the dam itself survived, but a huge **landslide** (~270 million m^3) fell into the reservoir from the slope.
  - Cause: bedding dipping **towards the reservoir**, clay layers, old slide mass, rise in reservoir level and heavy rain reduced shear strength.
  - Result: giant wave overtopped the dam, ≈ 2000 lives lost. **Lesson:** study reservoir-rim slope stability.

(b) St. Francis Dam, California, USA (1928)

- 62 m **concrete gravity dam** failed within hours of first full filling; ≈ 430 people died.
  - Cause: the left abutment had old landslide mass and **weak schist**; the right side had **conglomerate that softened when wet** (slaking, uplift).
  - **Lesson:** rock type and its behaviour in contact with water must be tested.

(c) Malpasset Dam, France (1959)

- Thin **arch dam** on gneiss; abutment had a hidden **fault / foliation surfaces**. Water pressure (uplift) pushed a rock wedge out - ≈ 420 deaths.
  - **Lesson:** discontinuities (joints, faults) in abutments govern arch dam safety.

(d) Teton Dam, Idaho, USA (1976)

- Earth-fill dam failed on first filling: **seepage through fissured volcanic rock** (rhyolite) and the key trench caused piping and a breach.
  - **Lesson:** grouting and cut-off for jointed foundation rock; monitor seepage.

(e) Koyna Dam (Maharashtra, India) - 1967

- Reservoir-induced earthquake (M 6.5) near a **gravity dam** on Deccan basalt - dam was damaged but did not collapse.
  - **Lesson:** reservoir loading can trigger seismicity (RIS); seismic design is essential.

(f) Konkan Railway & Himalayan roads (India)

- Konkan Railway (Mumbai-Mangalore): many tunnels and cuttings in laterite, basalt and soft ground; **geological mapping** reduced slope failures and tunnel collapses.

- Himalayan highways: landslides on weathered, sheared and jointed rocks each monsoon - slope-cutting without geology = repeated failures.

Project	Geological issue	Lesson
Vajont (1963)	Bedding slope towards reservoir	Check rim stability
St. Francis (1928)	Weak schist, soft conglomerate	Test rock in water
Malpasset (1959)	Fault + foliation at abutment	Map discontinuities
Teton (1976)	Seepage in fissured rock	Grouting / cut-off
Koyna (1967)	Reservoir-induced quake	Seismic design

2. Weathering of Rocks

Weathering = the in-situ (in-place) breakdown of rocks at or near the earth's surface by the action of atmosphere, water and organisms **without transportation of the material.**

- Weathering is **different from erosion**: erosion = weathering + removal/transport by agents (water, wind, ice).
- Result: hard rock -> loose fragments -> **soil** (regolith).
- Rock weathering profile: fresh rock -> partially weathered rock -> saprolite -> residual soil.

2.1 Types of Weathering

1. **Physical (mechanical / disintegration)** - rock breaks into pieces; no change in chemical composition.
2. **Chemical (decomposition)** - minerals are altered and new minerals formed.
3. **Biological (organic)** - plants, animals, man and micro-organisms help both of the above.

2.2 Physical Weathering Processes

- **Temperature changes (insolation weathering)**: day heating - night cooling -> repeated expansion/contraction -> cracks. Common in **deserts**.
- **Exfoliation (onion-skin weathering)**: outer shells peel off like onion layers; seen in granite domes.
- **Frost action / ice wedging**: water enters cracks, freezes, volume increases ≈ 9% ->

prises rock apart. Cold regions.

- **Salt crystallisation**: salts crystallise in pores and grow - splitting rock (arid and coastal zones).
- **Pressure release (unloading)**: erosion of overlying rock reduces confining pressure -> sheet joints form.
- **Wetting and drying (slaking)**: shales, clays swell and shrink -> flaking.
- **Abrasion by wind, water, ice** - sometimes grouped here.

2.3 Chemical Weathering Processes

- **Oxidation**: oxygen reacts with Fe-minerals -> iron oxides (rust-brown limonite, red hematite). Example: $4FeO + O_2 \rightarrow 2Fe_2O_3$.
- **Hydration**: minerals absorb water and swell - anhydrite -> gypsum; hematite -> limonite.
- **Hydrolysis**: reaction of minerals with water (H+ and OH-); **feldspar -> clay (kaolinite)**; the most important process on silicate rocks.
  - $2KAlSi_3O_8 + 2H_2O + CO_2 \rightarrow Al_2Si_2O_5(OH)_4 + K_2CO_3 + 4SiO_2$
- **Carbonation**: CO_2 dissolved in rain makes carbonic acid H_2CO_3; it dissolves **limestone** -> caves, sinkholes. $CaCO_3 + H_2CO_3 \rightarrow Ca(HCO_3)_2$
- **Solution**: direct dissolving of soluble minerals (rock salt, gypsum, calcite).
- **Reduction**: removal of oxygen under waterlogged conditions ($Fe^{3+} \rightarrow Fe^{2+}$), grey-green colours.

2.4 Biological Weathering

- **Roots** of trees grow in joints and wedge rock apart (root wedging).
- **Burrowing animals** (worms, termites, rodents) mix and expose fresh surface.
- **Lichens, bacteria, fungi** secrete organic acids -> decompose minerals (chelation).
- **Man** - mining, quarrying, blasting, agriculture, construction speed weathering.

2.5 Factors Controlling Weathering

- **Climate**: temperature and rainfall - fastest in hot-humid regions (chemical); frost in cold regions (physical).
- **Rock type & mineral composition**: quartz resists; olivine, calcite, feldspar weather quickly.
- **Structure**: joints, bedding, faults give water entry paths and larger surface area.

- **Topography / slope**: steep slopes remove debris (thin soil); gentle slopes keep it (deep weathering).
- **Vegetation, time & ground water.**

Physical	Chemical	Biological
Mechanical break-up	Change in composition	Organisms
Dry / cold climate	Warm and wet climate	Any climate with life
Gives angular debris	Gives clay, solutions	Humus, soil

2.6 Importance of Weathering in Civil Engineering

- Weathered rock has **low strength and high permeability** - lowers bearing capacity; foundations must go below the weathered zone.
- Depth of weathering controls excavation depth, slope angle and cost of tunnelling.
- Weathered stones are **unsuitable** as building stone / aggregate (reduced durability).
- Weathering produces **soils** - agriculture, clay for bricks and earth dams; **laterite** and **bauxite** are weathering products.
- Causes landslides, rock falls, and decay of monuments and concrete.

2.7 Grades of Weathering (engineering classification)

Grade	Description
I - Fresh	No visible sign of weathering; rock sound and ringing sound on hammer blow
II - Slightly	Discolouration along joints; strength almost same as fresh rock
III - Moderately	Less than half of rock decomposed; weak, hammer gives dull sound
IV - Highly	More than half decomposed; can be broken by hand
V - Completely	All rock turned to soil but original texture visible
VI - Residual soil	Fully soil, original rock texture destroyed

- Foundations of heavy structures should rest on **Grade I-II rock**; Grade IV-VI need removal, grouting or deep piles.

2.8 Soil Formation (Pedogenesis)

- **Soil** = weathered rock + organic matter + air + water. Formed by weathering over long time under climate, parent rock, relief, organisms and time (**CLORPT**).
- **Soil profile (horizons):** O (humus) -> **A** (top soil, leached) -> **B** (sub-soil, accumulation) -> **C** (weathered parent rock) -> **R** (bed rock).

- Indian soils: **alluvial** (river plains), **black cotton / regur** (Deccan basalt; swells, shrinks, bad for foundations), **red** (granite, gneiss), **laterite** (hot, humid), **desert** and **mountain** soils.
  - Anantapur region (A.P.): mostly **red soils & black cotton soils** over granite-gneiss and Cuddapah rocks.
- **Residual soils** keep the structure of the parent rock; **transported soils** are loose and heterogeneous.

2.9 Relative Resistance of Minerals to Weathering

- Least resistant -> **olivine, pyroxene (augite), Ca-plagioclase, calcite** (form first at high temperature, unstable at the surface).
- Moderate -> hornblende, biotite, orthoclase.
- Most resistant -> **muscovite, quartz** (stable, forms sand, quartzite).
- Rule (Goldich series): minerals which crystallise first from magma weather first.

Common weathering products

- Feldspar -> clay (kaolinite) | Mica -> clay + iron oxides | Limestone -> solution, caves | Basalt -> black cotton soil / laterite | Granite -> red soil, sand

3. Geological Agents

Geological agents are natural forces that change the earth's surface and interior. They work continuously but slowly ("the present is the key to the past" - **Uniformitarianism**, James Hutton).

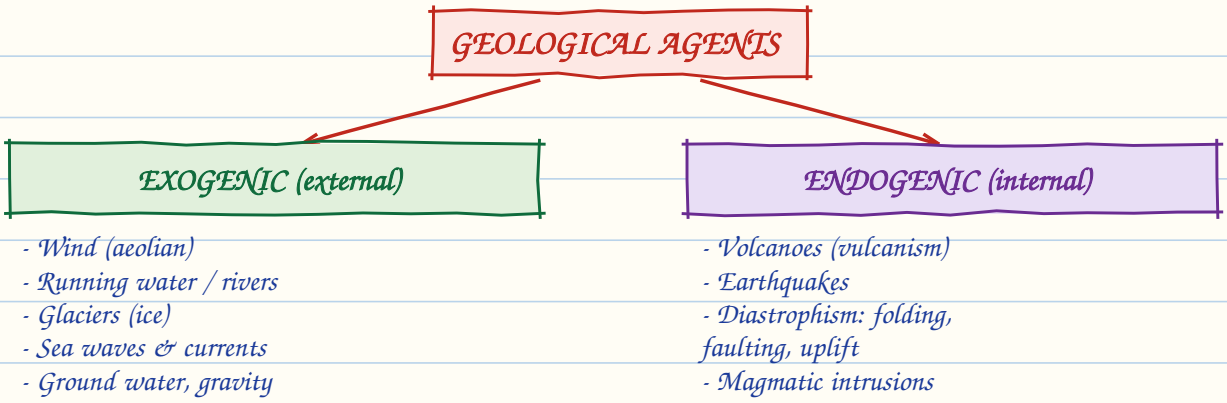


Fig 1: Geological agents

- **Gradation** = levelling of land by agents; has two parts: **degradation** (wearing down - weathering, erosion, transport) and **aggradation** (building up - deposition).
- Sequence of work of any external agent: **Weathering -> Erosion -> Transportation -> Deposition -> Compaction / Lithification.**

3.1 Work of the main agents (summary)

Agent	Erosional work	Typical landforms
River	Down-cutting, abrasion	V-valley, gorge, delta
Wind	Deflation, abrasion	Dunes, loess, mushroom rock
Glacier	Plucking, abrasion	U-valley, moraine, cirque
Sea waves	Hydraulic action	Cliffs, beaches, stacks
Ground water	Solution	Caves, sinkholes, stalactites

- **Engineering relevance**: agents decide soil type (alluvial, aeolian, glacial, marine), foundation problems, silting of reservoirs, coastal erosion and the shape of valleys used for dams.

4. Rivers and Geological Work of Rivers

A **river** is a natural stream of water flowing in a definite channel from higher to lower ground under gravity. **Rivers are the most powerful agent of denudation** on land, shaping valleys, plains and deltas.

- Parts of a river: **source (head)** -> **upper course** -> **middle course** -> **lower course** ->

mouth (sea / lake).

- **Tributaries** join the main stream; **drainage basin (catchment)** = area drained; **watershed / divide** separates two basins.
- Drainage patterns (seen on maps/aerial photos): **dendritic** (tree-like, uniform rock), **trellis** (parallel, folded strata), **radial** (volcano / dome), **rectangular** (joints/faults), **parallel** (steep uniform slope).

4.1 Erosion by Rivers

River erosion depends on **velocity, volume (discharge), slope, load carried and rock resistance.**

1. **Hydraulic action** - sheer force of moving water dislodges loose rock; compresses air in cracks.
2. **Abrasion (corrasion)** - rock fragments carried by water scrape the bed and banks like sandpaper; main erosional process; makes **potholes**.
3. **Attrition** - boulders collide and get **rounded and smaller** as they travel downstream.
4. **Solution (corrosion)** - soluble rock (limestone) dissolves in river water.

- Types of erosion by direction:
  - **Vertical (down-cutting)** - deepens valley in upper course -> V-shaped valley, gorge, canyon.
  - **Lateral (side-cutting)** - widens valley, forms meanders in middle/lower course.
  - **Headward erosion** - river extends upstream at its source; leads to **river capture (piracy)**.

Velocity facts

- Erosive power increases with the SIXTH power of velocity (v^6 law) - doubling velocity gives ≈ 64 times more load-carrying capacity! Floods therefore do most of the geological work.

4.2 Transportation by Rivers

- **Solution** - dissolved salts (Ca, Mg, Na, bicarbonates) carried invisibly.
- **Suspension** - clay, silt, fine sand held up by turbulence; makes river muddy.
- **Saltation** - sand grains hop along the bed in jumps.
- **Traction** - pebbles and boulders roll, slide, drag on the bed (**bed load**).
- **Competence** = largest particle a river can move; **Capacity** = total load a river can

carry.

4.3 Deposition by Rivers

River deposits load when **velocity falls**: entry into sea/lake, flattening of slope, reduction in discharge, widening of channel, or obstruction. Deposits are called **alluvium** (sand, silt, clay, gravel).

4.4 Landforms made by Rivers

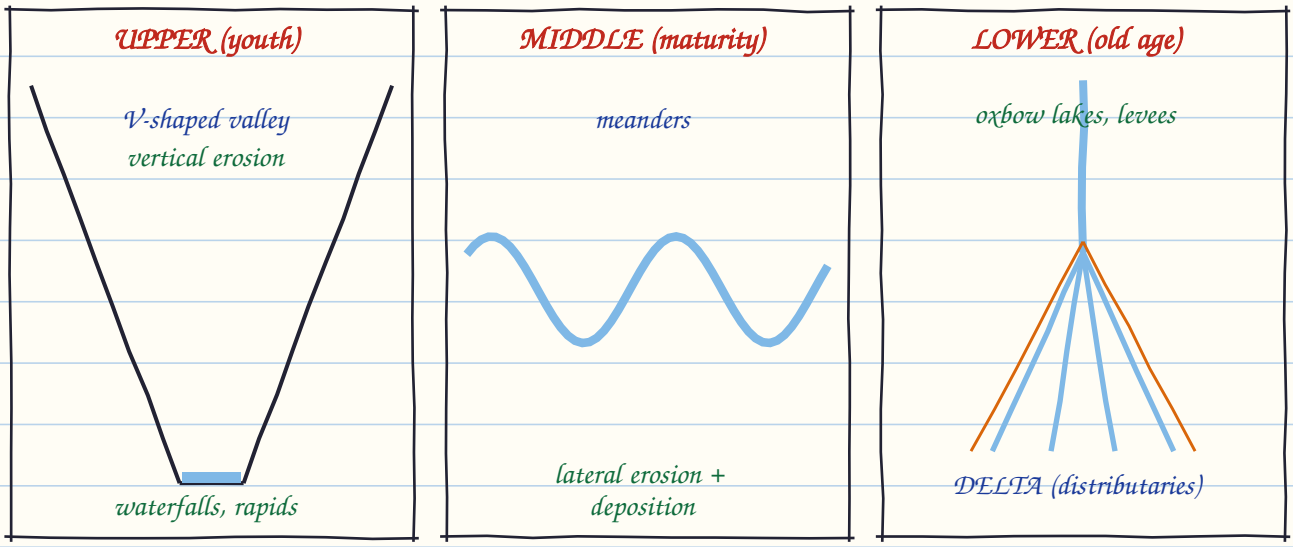


Fig 2: Three courses of a river and their landforms

Erosional landforms

- **V-shaped valley** - narrow, steep sides; typical of youthful stage.
- **Gorge & canyon** - deep narrow valley cut in hard rock (Grand Canyon; Gandikota gorge on River Penna, A.P.).
- **Waterfall & rapids** - hard rock band over soft rock; undercutting forms **plunge pool** (e.g., Jog Falls, Hogenakkal).
- **Potholes** - cylindrical holes drilled by swirling pebbles.
- **Interlocking spurs** - ridges alternate from the sides of a V-valley.

Depositional landforms

- **Alluvial fan / cone** - fan of debris at the foot of a mountain where slope suddenly decreases.
- **Meanders** - bends of river; erosion on **outer (concave) bank**, deposition on **inner (convex) bank** = **point bar**.
- **Oxbow lake** - a meander neck is cut off in flood, leaving a crescent-shaped lake.
- **Flood plain** - broad flat area built by flood deposits on either side of the channel

(fertile).

- **Natural levees** - raised banks of coarse sediment deposited at channel edge during floods.
- **Braided channel** - river splits into many interwoven channels around bars (heavy load, variable flow).
- **Delta** - triangular (Greek letter Δ) deposit at river mouth: Krishna-Godavari delta, Ganga-Brahmaputra delta (largest). **Types:** arcuate (Nile), bird-foot (Mississippi), estuarine.
- **River terraces** - step-like flat surfaces left when the river cuts down into its old flood plain (rejuvenation).

4.5 Stages (Cycle) of a River - W. M. Davis

Feature	Youth	Maturity	Old age
Valley	V-shaped, narrow	Wider, open	Very broad flood plain
Gradient	Steep	Moderate	Almost flat
Work	Vertical erosion	Erosion + deposition	Deposition mostly
Features	Falls, rapids, gorge	Meanders, tributaries	Oxbow, levees, delta

- **Rejuvenation** - renewed down-cutting due to uplift of land or fall in sea level -> terraces, incised meanders, knick-points.

4.6 Engineering Importance of River Work

- **Dam & reservoir sites** - narrow gorges in strong rock in the upper course (e.g., Nagarjuna Sagar on River Krishna). Wide flood plains are poor sites for dams.
- **Bridges** - foundation must resist **scour** (erosion around piers); choose straight, stable reaches; avoid meander bends.
- **River training works** - guide banks, spurs, embankments, bank protection by riprap to stop lateral erosion.
- **Flood control** - study of flood plains, discharge records and levees.
- **Siltation** - sediment load fills reservoirs and reduces life; check catchment erosion.
- **Construction materials** - river sand, gravel and boulders for concrete; **alluvial soils** are poor foundations (soft, compressible, high water-table).
- **Water supply** - alluvium is the best aquifer; infiltration wells in river beds.
- **Pipelines, canals and roads** - route along stable terraces and ridges; avoid active

channel shifting.

Exam points (5 marks)

- 1. Define weathering and list types. 2. Explain river erosion, transportation and deposition. 3. Name 5 landforms of rivers with sketches. 4. Write two case studies showing importance of geology.

4.7 Important Terms - Quick Revision

- **Weathering** = in-situ breakdown; **Erosion** = wearing away with removal; **Denudation** = both together; **Gradation** = levelling.
- **Regolith** - loose cover of weathered rock; **Soil** - regolith with organic matter that supports plants.
- **Residual soil** stays at place; **transported soil** (alluvial, aeolian, glacial, marine, colluvial) is carried by agents.
- **Hydraulic action, abrasion, attrition, solution** = the four erosion mechanisms of a river.
- **Meander** - erosion outside, deposition inside; **delta** - deposition at mouth; **levee** - deposition at edges.

5. Short Answer Revision (important questions)

1. **Define engineering geology.** Application of geological principles to investigation, design, construction and maintenance of civil engineering works.
2. **Why is weathering studied by civil engineers?** It reduces strength and increases permeability of rock; decides depth of foundation and quality of construction material.
3. **Difference between weathering and erosion?** Weathering is in-situ breakdown; erosion includes removal by moving agents.
4. **Name the three types of weathering.** Physical, chemical and biological.
5. **What is exfoliation?** Peeling of curved shells from rock surface due to temperature change / unloading.
6. **Which process converts feldspar into clay?** Hydrolysis.
7. **Name any four branches of geology.** Mineralogy, petrology, structural geology,

hydrogeology, geomorphology.

8. **What is a delta?** Fan or triangle-shaped alluvial deposit at river mouth when velocity falls.
9. **Define meander and oxbow lake.** Bend in a river; a cut-off abandoned meander loop.
10. **Why are dam sites chosen in the upper course?** Narrow valley, strong rock, large head and small dam length = low cost.

Last-minute tips

- - Always draw a neat labelled sketch for rivers (V-valley, meander, delta).
- - For case studies write: place - year - type of structure - geological cause - lesson.
- - Use words like in-situ, hydrolysis, abrasion, attrition, competence, aggradation.

ENGINEERING GEOLOGY

UNIT - II : MINERALOGY AND PETROLOGY

Minerals, Rocks - Igneous, Sedimentary & Metamorphic

Syllabus (as given)

- Definitions of mineral and rock - methods of study. Physical properties of minerals and rocks (megascopic). Rock forming minerals: Feldspar, Quartz, Olivine, Augite, Hornblende, Mica, Asbestos, Talc, Chlorite, Kyanite, Garnet, Calcite; ore minerals: Pyrite, Hematite, Magnetite, Galena, Pyrolusite, Graphite, Chromite, Bauxite. Classification, structures, textures and forms of Igneous, Sedimentary and Metamorphic rocks; megascopic study of granite varieties (pink, gray, green), Pegmatite, Dolerite, Basalt, Shale, Sandstone, Limestone, Laterite, Quartzite, Gneiss, Schist, Marble, Khondalite and Slate.

1. Mineral and Rock - Definitions

- **Mineral**: a naturally occurring, **inorganic, homogeneous solid** with a definite chemical composition (or narrow range) and an ordered **crystalline atomic structure**. Eg: quartz (SiO_2), calcite ($CaCO_3$).
- **Rock**: a natural **aggregate of one or more minerals** (or mineral matter) forming a part of the earth's crust. Eg: granite = quartz + feldspar + mica.
  - Monomineralic rock: limestone (calcite), quartzite (quartz), marble. Polymineralic: granite, gneiss.
- **Crystal**: a mineral with regular geometric shape bounded by plane faces; **Amorphous**: no regular structure (opal, volcanic glass).
- About **2 – 3 thousand minerals** are known, but only ~**12 rock-forming minerals** make up 99% of the crust; **Silicates** are 90%.
- Abundance of elements in the crust (weight %): **O 46.6, Si 27.7, Al 8.1, Fe 5.0, Ca 3.6, Na 2.8, K 2.6, Mg 2.1**.

1.1 Methods of Study of Minerals and Rocks

1. **Megascopic (hand specimen) study** - by naked eye / hand lens (10x): colour, luster, hardness, cleavage etc. - done in field and lab.
2. **Microscopic study** - thin section (0.03 mm) under **petrological (polarising) microscope** - optical properties, texture, mineral percentages.
3. **Chemical analysis** - wet analysis, atomic absorption, XRF.
4. **X-ray diffraction (XRD)** - identifies crystal structure; **SEM / EPMA** - micro-structure and composition.
5. **Blow-pipe analysis, thermal (DTA) & electron methods** - additional tests.

For Civil Engineers

- Megascopic study is the most useful: rocks and minerals are identified in the field and in the stone yard within minutes.

2. Physical Properties of Minerals (megascopic)

Property	Meaning / how to test
Colour	Appearance in reflected light; may vary with impurities (quartz: white, pink, purple). Use fresh surface.
Streak	Colour of mineral **powder** on unglazed porcelain plate. Hematite = cherry red; pyrite = greenish black.
Luster	Way light reflects: **metallic, sub-metallic, vitreous (glassy), resinous, pearly, silky, greasy, earthy, adamantine.**
Form / habit	Shape: prismatic, tabular, fibrous (asbestos), massive, granular, botryoidal, foliated, acicular.
Cleavage	Tendency to split along smooth flat planes: perfect, good, imperfect (mica - 1 direction, calcite - 3 directions not at 90°).
Fracture	Irregular break in non-cleavage direction: **conchoidal** (quartz), uneven, splintery, hackly, earthy.
Hardness	Resistance to scratching - **Mohs' scale** (1-10).
Specific gravity	Density relative to water: determined with Walker's steelyard / pycnometer. Metallic minerals have high SG.

Tenacity	Brittle, malleable, sectile, flexible, elastic (mica).
Other	Magnetism (magnetite), taste (halite), effervescence with dil. HCl (calcite), transparency, fluorescence, double refraction.

2.1 Mohs' Scale of Hardness

No.	Mineral	No.	Mineral	Field test
1	Talc	6	Orthoclase	Fingernail (2.5), copper coin (3.5)
2	Gypsum	7	Quartz	Knife blade / glass (5.5)
3	Calcite	8	Topaz	Steel file (6.5)
4	Fluorite	9	Corundum	Cuts glass
5	Apatite	10	Diamond	Hardest known

- Memory: **T**all **G**irls **C**an **F**latter **A**ll **O**ld **Q**uiet **T**all **C**ats **D**ear.

3. Common Rock-Forming Minerals

3.1 Feldspar Group (most abundant: ≈ 50-60% of crust)

- Alumino-silicates of **K, Na, Ca**. Two sub-groups:
  - **Orthoclase / Microcline (K-feldspar)** $KAlSi_3O_8$: colour white, pink, flesh; vitreous; **H = 6**; cleavage 2 directions ~90°; SG 2.56. Pink colour of granite.
  - **Plagioclase (Na-Ca)**: albite -> anorthite; white to grey; H = 6-6.5; cleavage 2 directions (86°); SG 2.6-2.76; shows **twin striations**.
- Uses: ceramics, glass, porcelain; weathers to **clay** (kaolin).

3.2 Quartz Group (SiO_2)

- **Varieties**: rock crystal (colourless), **milky** quartz, **rose** quartz (pink), **amethyst** (violet), smoky, citrine; cryptocrystalline: **chalcedony, agate, jasper, flint, chert**.
- Properties: vitreous luster; **H = 7**; SG 2.65; **no cleavage, conchoidal fracture**; hexagonal prism with pyramid; transparent to translucent.
- Uses: glass, abrasive, silicon chips, concrete aggregate; most resistant mineral to weathering -> **sand**.

3.3 Mica Group (sheet silicates)

- **Muscovite** (white mica) $KAl_2(AlSi_3O_{10})(OH)_2$: colourless / silvery; **H 2-2.5**; **perfect basal cleavage** -> thin, flexible, elastic sheets; SG 2.8-3.
- **Biotite** (black mica): $K(Mg,Fe)_3(AlSi_3O_{10})(OH)_2$: brown-black, pearly; same cleavage; SG 2.7-3.1.
- Uses: **electrical insulators**, paints, cosmetics; decays to clay and iron oxides.

Mineral	Colour / Luster	H	Cleavage / Form	SG	Remarks & uses
Olivine $(Mg,Fe)_2SiO_4$	Olive green; vitreous	6.5-7	None; conchoidal; granular	3.3	Peridot gem; ultra-basic rocks; weathers fast
Augite (pyroxene)	Dark green-black; vitreous	5-6	2 sets at ~87-93°; short prism	3.3	In basalt, gabbro; 8-sided cross section
Hornblende (amphibole)	Black-dark green; vitreous	5-6	2 sets at 56° & 124°; long prism	3.2	In granite, diorite, gneiss; 6-sided section
Asbestos (fibrous amphibole / serpentine)	White-grey-green; silky	5-6	Fibrous, flexible	2.5-3	Fire-proof, insulation; **health hazard**
Talc (hydrous Mg silicate)	White-pale green; pearly, greasy	1	Foliated, soapy feel	2.7	Talcum powder, soapstone, paint
Chlorite	Green; pearly	2-2.5	One perfect cleavage; flakes **flexible not elastic**	2.6-3	Low-grade metamorphic rocks (schist)
Kyanite Al_2SiO_5	Blue; vitreous-pearly	4-5 along / 6-7 across	One perfect + one good; bladed	3.6	Refractory; indicates **high-pressure metamorphism**
Garnet	Red-brown; vitreous-resinous	6.5-7.5	None; dodecahedral crystals	3.5-4.3	Gem, abrasive; in mica schist, gneiss

Calcite $CaCO_3$	White/colourless; vitreous	3	**Perfect 3 directions (rhombohedral)**	2.7	Fizzes with cold dil. HCl; cement, lime, marble

Quick identification clues

- Olivine = green, no cleavage. Hornblende vs Augite: 124° vs 87° cleavage angle. Kyanite = blue, two hardnesses. Calcite = effervescence. Talc = softest.

4. Ore Forming (Economic) Minerals

Mineral	Composition / Colour	H	Streak / Luster	SG	Uses / Occurrence
Pyrite "fool's gold"	FeS_2; brass yellow	6-6.5	Greenish-black; metallic	5	Sulphur, sulphuric acid; cubes striated
Hematite	Fe_2O_3; steel-grey to red	5.5-6.5	**Cherry red**; metallic	5.2	Principal **iron ore**; red pigment
Magnetite	Fe_3O_4; iron-black	5.5-6.5	Black; metallic	5.2	Rich iron ore; **strongly magnetic**
Galena	PbS; lead-grey	2.5	Lead-grey; metallic	7.5	Ore of **lead**; cubic cleavage
Pyrolusite	MnO_2; black-grey	2-2.5	Black; sub-metallic, soils fingers	4.8	Main **manganese** ore; steel, batteries
Graphite	C; black-steel grey	1-2	Black; greasy, metallic	2.2	Pencils, lubricant, electrodes; soft, greasy
Chromite	$FeCr_2O_4$; black	5.5	Brown; sub-metallic	4.5	Only ore of **chromium**; stainless steel
Bauxite	$Al_2O_3.nH_2O$; white-red-brown, earthy	1-3	White; earthy, pisolitic	2.5	Ore of **aluminium**; laterite product

- India: **iron** (Odisha, Jharkhand, Karnataka), **bauxite** (Odisha, A.P.-Araku), **manganese** (Odisha, Maharashtra), **chromite** (Sukinda, Odisha), **lead-zinc** (Rajasthan,

Zawar), **mica** (Nellore, A.P.).

5. Petrology - Study of Rocks

Rocks are of **three genetic types**; all are linked by the **rock cycle**.

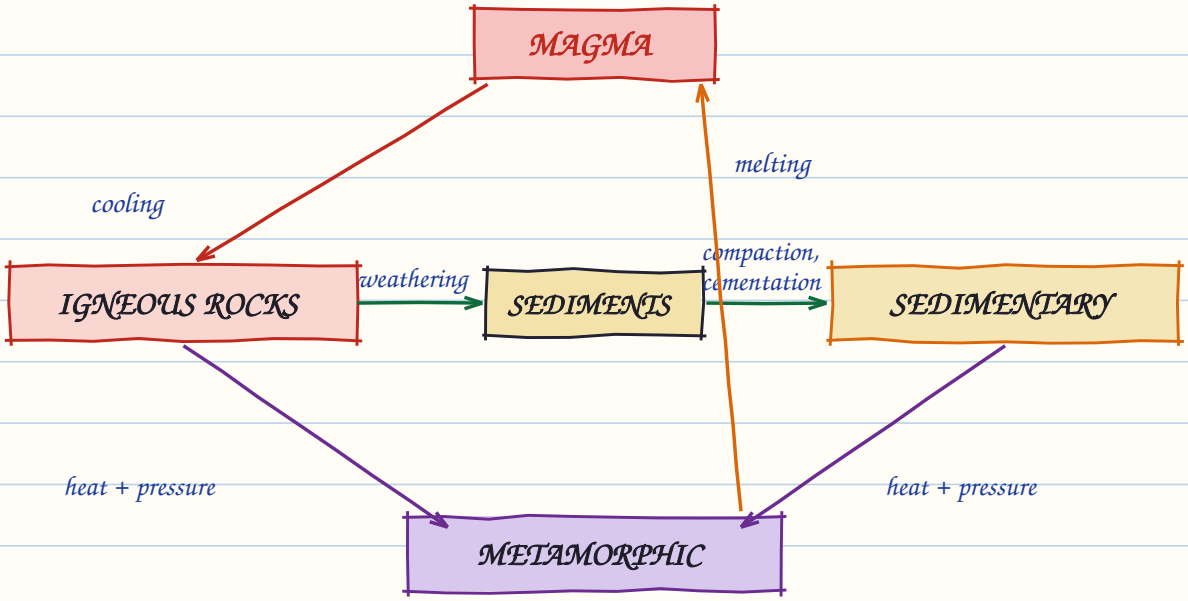


Fig 1: The Rock Cycle

5.1 Igneous Rocks (Primary / parent rocks)

- Formed by **cooling and solidification of magma (molten rock below surface) or lava (on surface)**. Make up ≈ 95% of the crust (volume).
- **Classification by place of consolidation:**
  - **Plutonic (abyssal)** - deep, very slow cooling, **coarse-grained** (granite, gabbro, diorite).
  - **Hypabyssal (dyke / sill)** - intermediate depth, medium grain, porphyritic (dolerite, pegmatite).
  - **Volcanic (extrusive)** - surface, rapid cooling, **fine-grained or glassy** (basalt, rhyolite, obsidian).
- **Classification by silica content:**

Type	SiO_2 %	Minerals	Examples
Acid	> 65	Quartz, K-feldspar, mica; light colour	Granite, rhyolite
Intermediate	55-65	Plagioclase, hornblende	Diorite, andesite

Basic	45-55	Ca-plagioclase, augite; dark	Gabbro, dolerite, basalt
Ultrabasic	< 45	Olivine, pyroxene; very dark	Peridotite, dunite

Textures of Igneous Rocks (size, shape, arrangement of grains)

- **Phaneritic** - grains visible to naked eye (granite); **Aphanitic** - too fine to see (basalt); **Glassy** - no crystals (obsidian).
- **Porphyritic** - large crystals (**phenocrysts**) in a fine **groundmass** (two stages of cooling).
- **Equigranular / granular** - grains of equal size; **Pegmatitic** - very large crystals (> 1 cm); **Ophitic** - laths of plagioclase enclosed in augite (dolerite).
- **Vesicular** - gas holes; **Amygdaloidal** - holes filled with calcite/quartz.

Structures of Igneous Rocks

- **Vesicular, amygdaloidal, ropy / pahoehoe** and **pillow** (underwater lava), **flow banding**, **columnar jointing** (basalt), **sheet joints, xenoliths**.

Forms of Igneous Bodies

(dyke cuts bedding; sill is parallel to bedding)

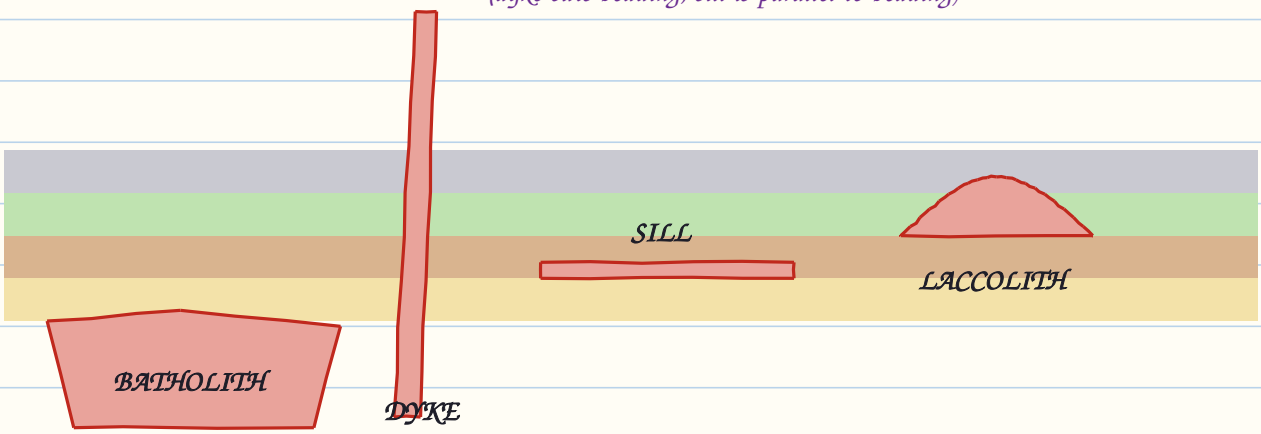


Fig 2: Forms of igneous intrusions

- **Intrusive:** batholith (largest, granite), **stock, laccolith** (lens-shaped dome), **lopolith** (saucer), **sill** (sheet parallel to beds), **dyke** (sheet cutting across beds), **volcanic neck**.
- **Extrusive:** lava flows, **lava plateau (Deccan traps)**, volcanic cones, pyroclastics (ash, bombs).

5.2 Megascopic Study of Igneous Rocks

Rock	Megascopic description	Engineering use

Granite - pink	Coarse, **pink orthoclase** + quartz + biotite; hard, dense, SG 2.6-2.7, compressive strength 100-250 MPa	Dimension stone, monuments, **dam & bridge foundations**, aggregate
Granite - grey	Grey plagioclase / quartz + biotite and hornblende; uniform	Building stone, paving, **polished slabs**
Granite - green	Greenish feldspar (amazonite / altered, chloritised)	Ornamental / cladding
Pegmatite	**Very coarse** crystals (cm to m) of quartz, feldspar, mica; occurs as veins	Source of mica, feldspar, gems; **weak zones** for foundations
Dolerite	Dark, medium-grained, **ophitic**; plagioclase + augite; SG 2.9	**Excellent road metal**, ballast, concrete aggregate; occurs as dykes
Basalt	Black-dark grey, **fine-grained**, may be vesicular or amygdaloidal; columnar joints; SG 2.9-3.0	Aggregate, road metal, **dams (Deccan)**; weathers to black-cotton soil

5.3 Sedimentary Rocks (Secondary rocks)

- Formed by **weathering -> erosion -> transport -> deposition -> lithification** (compaction + cementation) of older rocks, chemical precipitation or organic remains. Cover ≈ 75% of land surface but only ≈ 5% of crust volume.
- **Classification:**

Class	Origin	Examples
Mechanical / clastic (detrital)	Fragments of rocks	Conglomerate, **sandstone, shale**, breccia
Chemical	Precipitation from solution	**Limestone**, rock salt, gypsum
Organic	Remains of animals / plants	Coal, shell limestone, chalk
Residual	In-situ weathering residue	**Laterite**, bauxite

- Grain size (Wentworth): boulder > cobble > pebble (> 2 mm) -> **sand (2-1/16 mm)** -> **silt (1/16-1/256)** -> **clay (< 1/256 mm)**.

Structures (primary) of sedimentary rocks

- **Stratification / bedding** (layers, planes of deposition), **lamination** (< 1 cm), **cross-bedding** (inclined layers in dunes/rivers), **graded bedding**, **ripple marks**, **mud**

cracks, rain prints, fossils, concretions, nodules.

- Texture: clastic (rounded / angular grains), crystalline (chemical), bioclastic; cement = silica, calcite, iron oxide.

5.4 Megascopic Study of Sedimentary Rocks

Rock	Megascopic description	Engineering use
Shale	Very fine, laminated, splits into thin sheets (fissile); grey-black-red; soft, absorbs water, slakes	Raw material for bricks, cement; poor foundation / unstable slope
Sandstone	Gritty feel; sand grains of quartz cemented by silica / calcite / iron oxide; cross-bedding; colour white, buff, red	Building and paving stone, aggregate; silica-cemented = strong, ferruginous weaker
Limestone	Calcite; grey-white; fine to crystalline; effervescence with dil. HCl; may have fossils; H = 3	Cement, lime, flux, building stone; solution cavities (karst) - leakage in reservoirs
Laterite	Reddish-brown porous rock, rich in Fe and Al oxides; soft when quarried, hardens on exposure; pitted surface	Building blocks (Kerala, Konkan), road metal; bauxite in some

5.5 Metamorphic Rocks (Transformed rocks)

- Metamorphism = change of mineral composition / texture / structure of pre-existing rocks in the solid state due to heat, pressure and chemically active fluids, without melting.
- Types: Contact (thermal) - near igneous intrusion, heat only (marble, hornfels); Regional (dynamothermal) - large areas, heat + pressure (schist, gneiss); Dynamic (cataclastic) - along faults, pressure (mylonite); Hydrothermal / metasomatism.
- Grades: low (slate) -> medium (phyllite, schist) -> high (gneiss, granulite).
- Index minerals: chlorite -> biotite -> garnet -> kyanite -> sillimanite (increasing grade).

Textures and Structures

- Foliated (layered): slaty cleavage, schistosity, gneissose banding. Non-foliated: granulose, granoblastic (marble, quartzite), hornfelsic.

- **Porphyroblastic** (large new crystals in fine matrix) - garnet schist.
- Structures: **slaty cleavage, schistosity, gneissic banding, lineation, augen.**

Foliation order (increasing grade)

- Slate (very fine, rings when struck) -> Phyllite (silky sheen) -> Schist (visible flaky mica, wavy) -> Gneiss (coarse light and dark bands)

5.6 Megascopic Study of Metamorphic Rocks

Rock	Parent rock	Megascopic description	Engineering use
Quartzite	Sandstone	Very hard, **interlocking quartz** grains, glassy, no foliation, conchoidal fracture; white-pink	Road metal, aggregate, railway ballast; **strong dam foundations**
Gneiss	Granite / sediments	Coarse, **banded** - light (quartz-feldspar) and dark (mica, hornblende) layers	Building stone, paving; **good foundations** (planes of weakness along banding)
Schist	Shale / basalt	Medium-coarse, **wavy flaky** minerals (mica, chlorite, talc, garnet); splits readily	**Poor**: weak, slippery along schistosity - slope / tunnel problems
Marble	Limestone / dolomite	Crystalline, saccharoidal, **effervescence with HCl**, white or coloured veins; H = 3-4	Monuments, flooring, statues, decorative stone
Khondalite	Aluminous (pelitic) sediments	Granulite-grade, **garnet + sillimanite + quartz + graphite**, coarse banded; abundant in **Eastern Ghats (Andhra, Odisha)**	Local building stone; weathers to red soil
Slate	Shale / clay	**Very fine**, perfect slaty cleavage - splits into thin flat slabs; grey-black; rings when struck	Roofing, flooring, blackboards, damp-proof course

5.6A Additional Notes - Origin and Field Identification

Silicate mineral structure (why silicates dominate)

- Basic unit = **SiO_4 tetrahedron**. Linking pattern gives groups: **isolated** (olivine, garnet) -> **single chain** (pyroxene / augite) -> **double chain** (amphibole / hornblende) -> **sheet**

(mica, talc, chlorite) -> **framework** (quartz, feldspar).

- More linking = **harder, more stable** mineral: that is why quartz is the last to weather.
- **Bowen's reaction series:** first to crystallise (olivine, Ca-plagioclase) -> last (muscovite, quartz). **Discontinuous** (Fe-Mg minerals) and **continuous** (plagioclase) branches.

Formation of Laterite and Bauxite

- **Laterite**: intense **chemical weathering** of basalt / granite / khondalite in **hot, humid, alternating wet-dry climate** - silica and bases are leached out; **Fe and Al oxides** remain. Found in **Kerala, Karnataka coast, Odisha, East Godavari (A.P.)**.
- **Bauxite**: laterite rich in **alumina**, mined for aluminium (Araku Valley, Visakhapatnam).

Common Field Tests

- **HCl test:** calcite / limestone / marble fizz; **Scratch test:** knife (5.5), nail (2.5); **Streak test:** hematite red.
- **Magnet test:** magnetite; **Lustre:** pyrite metallic; **Grain size:** hand lens.
- **Colour index** of igneous rocks = % dark minerals: **leucocratic** (light), **mesocratic**, **melanocratic** (dark).

Rock Identification Flow

Question	If YES	If NO
Is it layered / banded / flaky?	Metamorphic (slate, schist, gneiss) or sedimentary (shale)	Go to next
Are crystals interlocking and visible?	Igneous coarse (granite, gabbro) or marble / quartzite	Go to next
Is it very fine, dark, vesicular?	Basalt (igneous volcanic)	Clastic or chemical rock
Is it made of rounded / angular grains with cement?	Sandstone / conglomerate	Limestone / laterite

- **Colour clues:** pink granite -> K-feldspar; grey granite -> plagioclase; green granite -> chloritised or amazonite feldspar; **black** -> basalt / dolerite; **red-brown, porous** -> laterite; **white, fizzing** -> limestone / marble.

5.7 Engineering Properties - Quick Comparison

- **Igneous (granite, basalt, dolerite):** strong, massive, low porosity - best for foundations, aggregate, dam sites. Watch for **joints, dykes, weathered zones**.
- **Sedimentary:** variable. **Sandstone** good if well cemented; **shale** weak and swells; **limestone** has cavities and solution channels.
- **Metamorphic:** quartzite, gneiss, marble strong; **schist, slate, phyllite** weak along foliation planes - failures in cuts and tunnels.

Rock	Compressive strength (MPa)	Porosity (%)	Remarks
Granite	100-250	< 1	Best foundation rock
Basalt	100-300	0.1-1	Sound; fractured zones leak
Sandstone	20-170	5-25	Variable by cement
Limestone	30-250	1-15	Solution cavities
Shale	5-100	5-30	Weak; slakes
Quartzite	150-300	< 1	Very strong
Marble	50-150	< 2	Good, moderate
Slate	50-180	< 1	Anisotropic

Exam checklist

- Write definition + classification table + 3 textures/structures + engineering uses. Draw neat sketches of the rock cycle and igneous forms (dyke, sill, laccolith, batholith).

5.8 Short Revision Questions

1. **Why is quartz found as sand?** It is hard, chemically stable and resists weathering.
2. **Mineral with cleavage in 3 directions not at 90°?** Calcite (rhombohedral cleavage).
3. **Mineral with H = 1 and soapy feel?** Talc.
4. **Name the ore of aluminium, iron, lead, manganese and chromium.** Bauxite, hematite / magnetite, galena, pyrolusite, chromite.
5. **Which rock is a good road metal?** Dolerite, basalt, quartzite, granite.
6. **Define porphyritic texture.** Large phenocrysts in a fine groundmass.
7. **What are the parent rocks of marble, quartzite and slate?** Limestone, sandstone,

shale.

ENGINEERING GEOLOGY

UNIT - III : STRUCTURAL GEOLOGY

Strike, Dip, Outcrops, Folds, Faults, Joints & Unconformities

Syllabus (as given)

- Strike, Dip and Outcrop study of common geological structures associated with rocks such as Folds, Faults, Joints and Unconformities - parts, types, mechanism and their importance in Civil Engineering.

1. Introduction to Structural Geology

Structural Geology deals with the **geometry, origin and attitude of rock structures** produced by deformation (folding, faulting, jointing) of the earth's crust.

- Rocks are deformed by **stress** (compression, tension, shear) generated by plate movements. **Strain** = change in shape / volume. Rock behaviour: **elastic -> ductile (plastic) -> brittle (rupture).**
- **Folds** = ductile deformation; **faults and joints** = brittle deformation.
- Why study? Structures control the **stability of dams, tunnels, slopes, foundations and reservoirs**, and the movement of ground water.

1.1 Attitude of Beds - Strike, Dip and Outcrop

- **Bed (stratum)** = layer of sedimentary rock bounded by bedding planes; **bedding plane** - surface of separation. The orientation of a bed in space is its **attitude**, specified by **strike and dip**.

Strike is 90° to Dip

Strike = trend of a level line
Dip = max. slope of bed
Tool: Brunton / clino compass
e.g. N 40° E / 30° SE

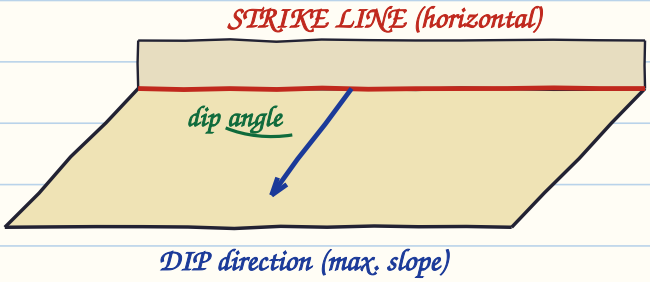


Fig 1: Strike and Dip of an inclined bed

- **Strike:** the direction of the **horizontal line** drawn on the surface of an inclined bed (a line of intersection of bed with a horizontal plane). Given as a **compass bearing**, e.g., N 40° E.
- **True dip:** the **maximum angle of inclination** of the bed with the horizontal, measured **perpendicular to strike**; has both **amount (angle)** and **direction** (e.g., 30° towards SE).
- **Apparent dip:** inclination measured in any direction **other than the true dip direction**; always **less than true dip**.
  - Formula: tan(apparent dip) = tan(true dip) × sin(angle between strike and the section line).
- **Instruments: Brunton / Clar compass** (strike, dip), clinometer.
- Notation on maps: a **long line** shows strike, a **short tick** shows dip direction with the dip angle written beside it. **Horizontal bed** = a circle with a cross (+) inside; **vertical bed** = strike line with a bar across it.

Outcrop

- **Outcrop** = the portion of a rock formation that is **exposed at the earth's surface** (not covered by soil). **Inlier / outlier** are older / younger exposures surrounded by the other.
- **Width of outcrop** depends on: (i) **thickness** of bed, (ii) **dip** of bed, (iii) **topography** (slope of the ground).

Bed attitude	Outcrop pattern on a map
Horizontal bed	Follows **contour lines** (outcrop parallel to contours) - dendritic on dissected terrain
Vertical bed	**Straight line**, not affected by topography; width = thickness
Inclined bed (on plane ground)	Straight belt; width = thickness / sin(dip)

Inclined bed across valley	**"V" shape - Rule of Vs:** V points **in the direction of dip** (beds dipping upstream, or downstream steeper than stream gradient); **exception:** beds dipping downstream at an angle **less than** the stream gradient - V points upstream (against dip)

- **Outcrop width:** steep dip -> **narrow outcrop**; gentle dip -> **wide outcrop**. Strata with the same dip but different thickness show different outcrop widths.
- **Uses of outcrop study:** geological mapping, finding structure (fold/fault), estimating **thickness** and **depth** of beds, locating dam sites and quarries.

True thickness

- $t = w \times \sin(\text{dip})$ on level ground (w = outcrop width). Beds dipping steeply give narrow bands - good for estimating tunnel and slope behaviour.

2. Folds

Fold = a **bend or wave-like curvature of rock layers** formed by plastic deformation due to **compressive stress**, mostly in sedimentary and metamorphic rocks.

2.1 Parts of a Fold

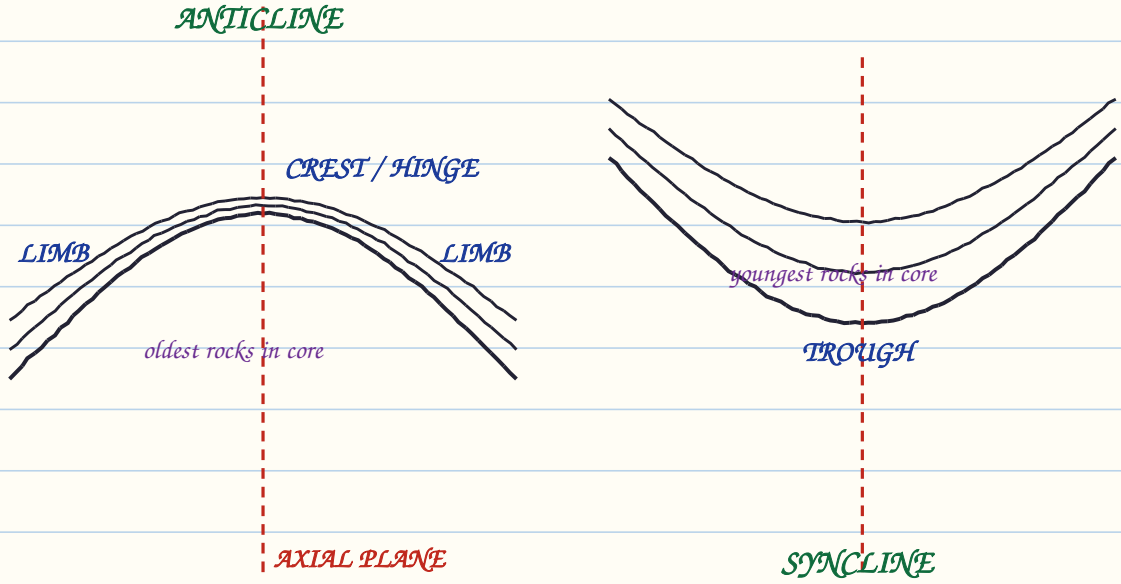


Fig 2: Anticline and syncline with parts

- **Limbs (flanks):** the two sides of a fold. **Hinge (crest/trough line):** line joining points of maximum curvature.
- **Axial plane:** imaginary plane dividing the fold into two halves (as symmetrically as

possible); **axis** = line where the axial plane meets a bed.

- **Crest** (top of anticline), **trough** (lowest part of syncline); **plunge** = inclination of the fold axis with the horizontal; **amplitude** (height), **wavelength**.
- **Inter-limb angle** - angle between the limbs: gentle (> 120°), open, close, tight, isoclinal (0°).

2.2 Types of Folds

Fold	Description
Anticline	Arch-shaped fold: limbs **dip away** from the axis; **oldest beds in the core**
Syncline	Trough-shaped: limbs **dip towards** the axis; **youngest beds in the core**
Monocline	Step-like bend; beds dip in one direction between horizontal strata
Symmetrical	Axial plane vertical, both limbs dip at equal angles
Asymmetrical	Axial plane inclined, limbs dip with unequal angles
Overturned	One limb is **inverted** (both dip in same direction)
Recumbent	Axial plane **horizontal**; fold lies on its side (nappe, Alps / Himalaya)
Isoclinal	Limbs **parallel** to each other, equal dip
Plunging	Axis inclined (not horizontal) - outcrop pattern is V-shaped / horse-shoe
Dome / basin	Beds dip outward in all directions (dome) / inwards (basin) - circular outcrop
Open / close / tight	Based on inter-limb angle

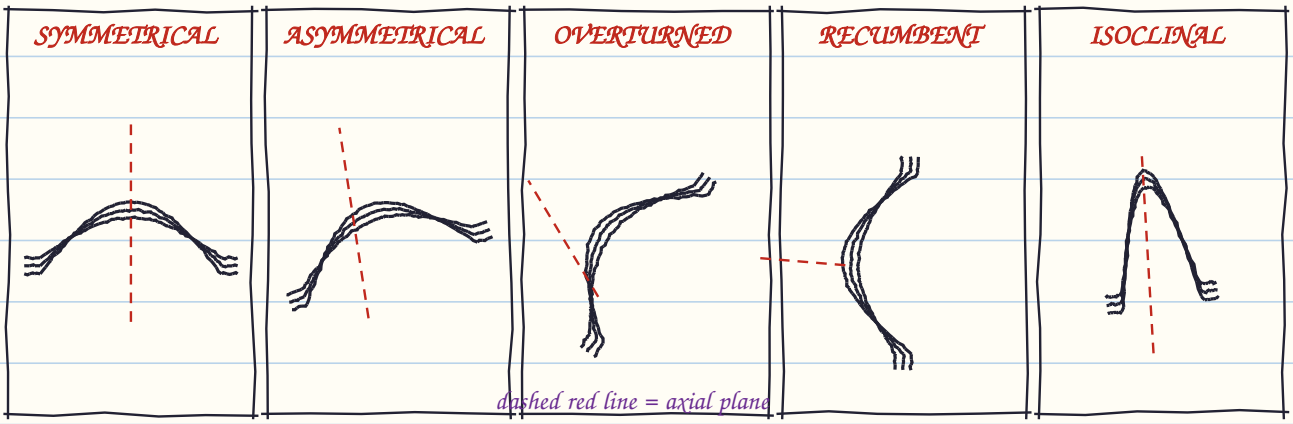


Fig 3: Fold types based on position of the axial plane

2.3 Mechanism of Folding

- **Flexural (buckling) folding:** layers **slip over each other** (like bending a pack of cards) under compression parallel to the beds.
- **Flow folding:** ductile layers flow by plastic deformation - common in rocks at depth, **high temperature**.
- **Shear (slip) folding:** movement along closely spaced shear planes, layers retain thickness measured parallel to axial plane.
- **Drape folding:** over a rising basement block (monocline).
- Controls: type and thickness of rocks, **temperature, confining pressure, rate of strain.**

2.4 Importance of Folds in Civil Engineering

- **Anticline** - tension on the crest -> open **joints, cracks**; suitable for ground water **only in limbs**; unsuitable for **reservoirs** (water leaks along fractures); good for **oil & gas** traps (crest).
- **Syncline** - compression on the trough; **good reservoir sites**; ground water accumulates (**artesian wells** in synclines).
- **Tunnels:** in a **syncline** the core rock is under compression (tight, stable) but **water collects** there - heavy inflow; in an **anticline** the crest has **tension cracks and open joints** (roof falls, over-break) but water drains away. Choose lining and drainage accordingly.
- **Dam sites: dip of beds upstream** (limbs dipping towards the reservoir) is preferred; dip towards downstream = **risk of sliding and seepage**.
- **Slopes:** beds dipping **parallel to slope** and steeper than friction angle -> **slides** (Vajont-type).
- **Highway cuttings:** avoid cutting across the dip direction of weak beds.
- Folded rocks give **trend of strata** - used for locating coal, ore bodies and aquifers.

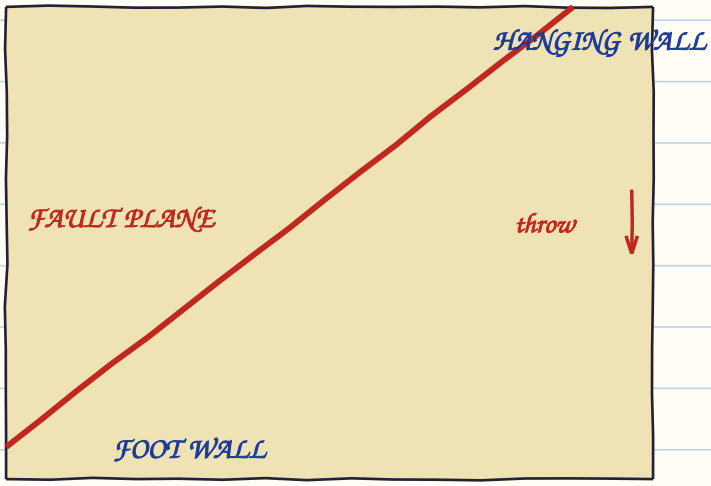
3. Faults

Fault = a **fracture in rocks along which there has been displacement** of the two sides relative to each other. Result of **brittle failure** when stress exceeds rock strength.

- Size: from centimetres to hundreds of km (San Andreas ~1200 km; Narmada-Son

lineament, Godavari rift in India). Movement may be sudden (**earthquake**) or slow (**creep**).

3.1 Parts / Terminology of a Fault



H.W. = block above fault plane
F.W. = block below the plane
THROW = vertical displacement
HEAVE = horizontal displacement
SLIP = true movement on plane
Fault dip = angle of plane

Fig 4: Parts of a fault

- **Fault plane** - surface along which the movement took place; may be vertical, inclined or horizontal; **fault zone** - broken zone with **gouge, breccia, mylonite**.
- **Hanging wall (HW)** - block **above** the inclined fault plane; **Foot wall (FW)** - block **below** it.
- **Dip of fault** - angle between fault plane and horizontal; **Strike of fault** - trend of the fault plane's horizontal line.
- **Throw** - vertical displacement; **Heave** - horizontal displacement; **Slip (net slip)** - actual relative displacement; **Hade** = 90° - dip.
- **Fault scarp** - cliff on the surface; **slickenside** - polished, striated surface; **drag folds** - bending of beds near the fault.

3.2 Classification of Faults

- **By relative movement (most common):**

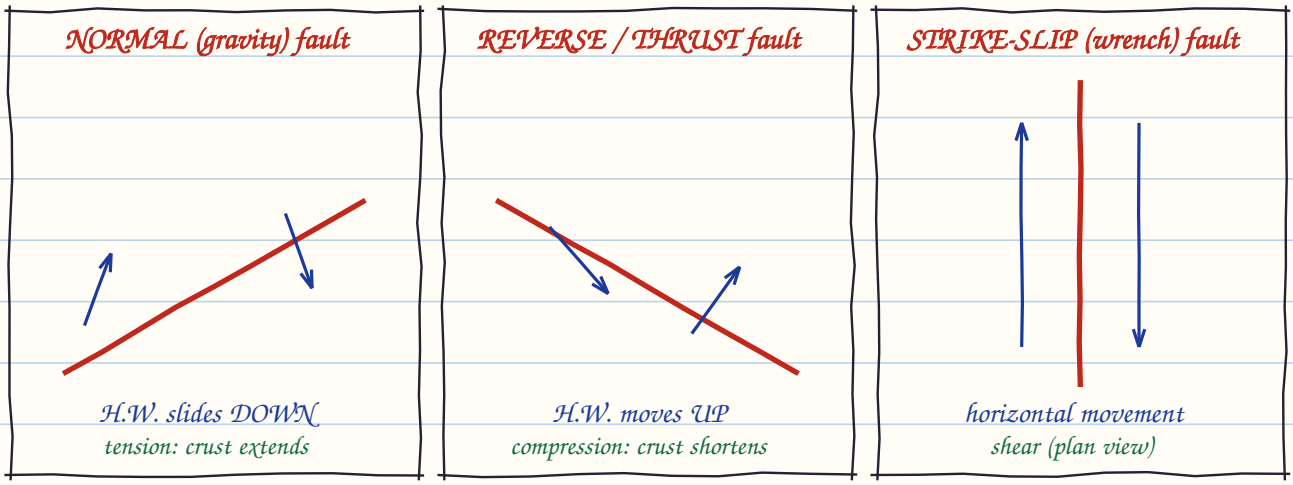


Fig 5: Normal, reverse and strike-slip faults

Type	Features
Normal (gravity) fault	Hanging wall moves DOWN; fault plane steep (60-70°); caused by tension (crustal extension); beds gap on the surface. Forms graben and horst
Reverse fault	Hanging wall moves UP relative to foot wall; caused by compression; beds repeat on the surface; Thrust = reverse fault with low dip (< 45°)
Strike-slip (wrench) fault	Movement horizontal along strike; dextral (right-lateral) or sinistral (left-lateral); vertical plane
Oblique-slip fault	Combination of dip-slip and strike-slip movement
Hinge / rotational fault	One block rotates about a point

- **By angle of dip:** high-angle (> 45°) and low-angle (< 45°).
- **By relation to strike of beds: strike fault** (parallel to strike), **dip fault** (parallel to dip), **oblique fault**.
- **Compound structures: Graben** = down-dropped block between two normal faults (rift valley, e.g., **Godavari valley**); **Horst** = up-thrown block; **step faults** = successive normal faults in same direction; **Nappe / thrust sheet**.

3.3 Causes and Recognition of Faults

- **Causes:** tectonic stress (plate movement), cooling and contraction of earth, **isostatic adjustment**, volcanic activity, gravity sliding.
- **Recognition in the field: sudden break / repetition or omission of beds, fault scarps, slickensides, fault breccia & gouge, drag folds,** offset of streams / ridges, springs and

lines of vegetation, silicification and mineralised zones.

3.4 Importance of Faults in Civil Engineering

- **Dam sites:** faults are **unsuitable** - zone of weak, crushed rock: **leakage, differential settlement**, and possible **reactivation (earthquake)**. Fault is a major seismic risk.
- **Reservoirs:** fault planes act as **channels for water loss**; if reservoir is near an active fault, **induced seismicity** (Koyna).
- **Tunnels:** fault zones give **heavy water inflow, roof falls, squeezing ground** - need special support, grouting; alignment should cross faults at right angles.
- **Foundations & bridges:** avoid active faults; foundation in sheared rock has low bearing capacity.
- **Slopes & roads:** sheared rocks fail; fault zones are weathered deeply.
- **Ground water:** fault gouge may be **impermeable** (barrier) or fault breccia **permeable** (conduit) - springs along faults.
- **Mineral deposits:** faults control ore bodies, oil traps, thermal springs.

Dam & fault - remember

- A dam must NEVER straddle an active fault. If a fault is unavoidable, use flexible design (earth/rock-fill), excavate and treat the gouge (dental concrete, grouting) and design for seismic loads.

4. Joints

Joint = a **fracture or crack in rock with NO appreciable displacement** of the two sides. Joints are the **most common structure** in rocks.

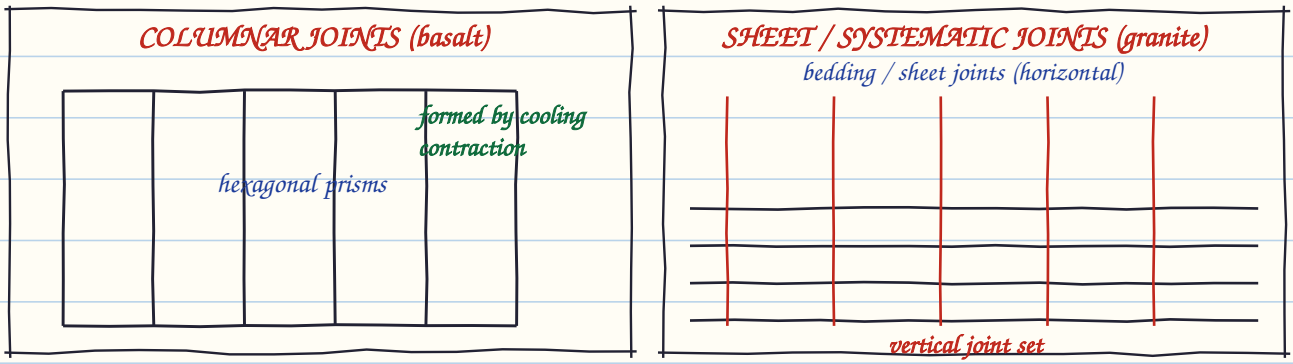


Fig 6: Types of joints

- **Parts:** joint plane, joint set (parallel joints), **joint system** (sets crossing - e.g.,

orthogonal), joint spacing, aperture, filling, persistence.

4.1 Types of Joints

- **Geometric classification: strike joint** (parallel to strike of beds), **dip joint** (parallel to dip), **oblique joint**, **bedding joint** (along bedding plane).
- **Genetic classification:**
  - **Tension joints** - formed by **stretching**, wide, rough, open - cut across rocks.
  - **Shear joints** - formed by shear stress; smooth, closed, occur in **conjugate sets** (about 60° between them).
  - **Columnar joints** - **polygonal (hexagonal) columns** from **contraction on cooling** of lava (basalt, Deccan traps).
  - **Sheet joints (exfoliation)** - curved, parallel to surface, from **unloading / pressure release** (granite domes).
  - **Mural joints** - cubical, in granite from cooling; **Flow joints, cleavage joints, release joints.**

4.2 Importance of Joints in Civil Engineering

- **Strength reduction:** joints are **planes of weakness** - rock mass strength is lower than intact rock strength; define **RQD** (Rock Quality Designation) and rock mass rating.
- **Permeability:** joints are paths for **water leakage** from reservoirs, seepage beneath dams; **grouting** is done to seal joints (curtain grouting).
- **Slope stability: wedge or planar failures** along joints dipping out of slope; rock bolts and anchors needed.
- **Tunnelling: joint orientation** controls roof falls, over-break; best when tunnel axis is **across** the strike of joint sets; **blasting** efficiency.
- **Quarrying:** joints help **block extraction** of dimension stone; closely jointed rock gives small blocks and **poor** building stone.
- **Foundation:** open joints lower bearing capacity; **weathering** proceeds through joints; **ground water** circulation.
- **Aquifers:** jointed granite/basalt forms **hard-rock aquifers** (bore wells in A.P.).

Feature	Fault	Joint	Fold
Nature	Fracture with displacement	Fracture, no displacement	Bend of layers

Stress	Tension / compression / shear	Tension / shear / release	Compression
Effect on dam	Highly dangerous	Needs grouting	Dip direction matters
Remedy	Avoid / treat gouge	Grouting, bolts	Select suitable limb

5. Unconformities

Unconformity = a **surface of erosion or non-deposition** separating younger strata from older rocks - it represents a **time gap (hiatus)** in the geological record. Beds above and below are **not in continuous sequence.**

- Formation steps: older rocks **deposited -> uplifted -> eroded** (or no deposition) -> **sea / land subsides** -> younger beds deposited above the eroded surface.

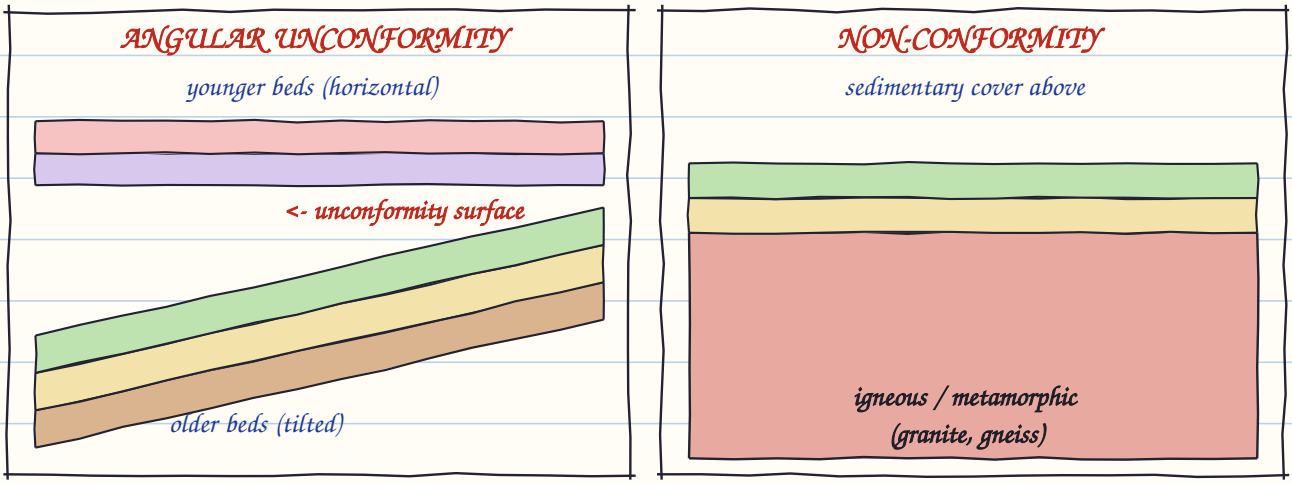


Fig 7: Angular unconformity and non-conformity

5.1 Types of Unconformities

Type	Description
Angular unconformity	Older beds are **tilted / folded**; younger beds above are **horizontal**; angle between them. Shows an earlier **orogeny** (mountain building)
Disconformity	Older and younger beds are **parallel** but a **surface of erosion** (irregular) separates them; evidence: channels, pebbles, soil layer
Non-conformity	**Sedimentary** beds rest on eroded **igneous or metamorphic** rocks (granite, gneiss) - plutonic rocks exposed by erosion

Paraconformity	Parallel beds, **no visible erosion**, only a gap in time (shown by fossils)
Local unconformity	Small-scale, due to local erosion / current action

- **Basal conglomerate** - a layer of pebbles at the base of the younger formation, strong evidence of unconformity.
- **Recognition:** (i) **discordance** of dip and strike, (ii) **basal conglomerate**, (iii) **missing fossil zones**, (iv) **weathered, irregular surface**, (v) abrupt change in rock type or metamorphism, (vi) **mapping** - outcrops of older beds truncated.
- **Overlap / Offlap**, **Outlier / Inlier** are related map patterns.

5.2 Importance of Unconformities in Civil Engineering

- The **unconformity plane** is usually a **weak plane**: weathered, porous and has **high permeability** - leakage in dam abutments and reservoirs; **sliding surface** for slopes.
- Foundations placed across an unconformity face **differential settlement** (hard rock below, soft strata above).
- **Tunnels** crossing an unconformity meet **abrupt change in rock** and possible **water inflow**.
- Indicate **ancient land surfaces** - help in mineral exploration (**bauxite, laterite, placer deposits, iron and manganese** are formed near them) and ground water.
- Basal conglomerate can be a **good aquifer**; also useful marker bed for correlation.

6. Summary - Structures and their Engineering Significance

Structure	Hazard	Engineering action
Anticline	Open joints at crest, leakage	Avoid reservoir on crest; grout
Syncline	Water accumulation	Good reservoir; tunnel drainage
Fault	Weak zone, seismic movement	Avoid / treat / flexible design
Joint	Seepage, rock wedge fall	Grouting, rock bolting, shotcrete
Unconformity	Weak interface, seepage	Excavate to sound rock; cut-off
Dip upstream	Safe	Preferred for dam sites
Dip downstream	Sliding & uplift	Anchorage, drainage, avoid

6.1 Field Measurement Tips (Notes for Lab / Viva)

- Hold the **Brunton compass level** on the bed surface, align edge along the **strike**, read bearing; then measure **dip** with clinometer, perpendicular to strike.
- Always record: **location, rock type, strike, dip amount & direction, thickness, structure** and sketch.
- Quote bearing in **quadrant** (N 30° W) or **azimuth** (330°) form; dip direction by compass quadrant.
- Distinguish **bedding, joint and cleavage** planes before measuring.

6.2 Short Revision Questions

1. **Define strike and dip.** Strike = direction of a horizontal line on the bed; dip = maximum slope angle measured at right angles to strike.
2. **Relation between strike and dip direction?** Always perpendicular (90°).
3. **Name the parts of a fold.** Limbs, crest, trough, axial plane, hinge, axis, plunge.
4. **Difference between anticline and syncline.** Anticline - upward arch, oldest beds in core; syncline - downward trough, youngest in core.
5. **Throw and heave?** Vertical and horizontal displacement across the fault.
6. **Which fault forms under tension?** Normal fault (graben).
7. **What is a basal conglomerate?** Coarse pebble bed at base of a younger formation resting on an unconformity.
8. **Why are faults avoided at dam sites?** Weak sheared zone, leakage and possible reactivation.

Exam tip

- Always draw: (1) strike-dip sketch, (2) anticline/syncline with parts, (3) normal / reverse / strike-slip faults, (4) angular unconformity. Label HW, FW, throw, heave, axial plane.

ENGINEERING GEOLOGY

UNIT - IV : EARTH HAZARDS & GEOPHYSICS

Ground Water, Earthquakes, Landslides & Geophysical Methods

Syllabus (as given)

- Ground Water: water table, cone of depression, geological controls, exploration techniques. Earthquakes and Landslides: terminology, classification, causes, effects, shield areas and seismic belts, Richter scale, intensity, precautions in seismic areas; classification, causes, effects and prevention of landslides. Geophysics: importance, classification, gravity, magnetic, electrical, seismic, radiometric methods; electrical resistivity, seismic refraction; engineering properties of rocks.

1. Ground Water

Ground water = water present in the **pores, cracks and voids of soils and rocks below the earth's surface**. It is the largest source of fresh water on land (after glaciers) and supplies > 60% of irrigation and drinking water in India.

- **Sources (origin): meteoric** (rain & snow infiltrating), **connate** (trapped in sediments during deposition), **juvenile / magmatic** (from magma), **recycled** (rivers, lakes, canals).
- **Hydrological cycle:** evaporation -> condensation -> precipitation -> infiltration / run-off -> ground water -> springs and rivers -> sea.

1.1 Important Terms

- **Porosity (n)** = volume of voids / total volume × 100 - ability to **store** water. Clay 45-55%, sand 30-40%, sandstone 5-25%, granite < 1%.
- **Permeability (K)** = ability to **transmit** water. Gravel and coarse sand high; clay and shale (high porosity but **very low permeability**).
- **Specific yield** = water that drains by gravity; **Specific retention** = water held by molecular forces. Porosity = Sy + Sr.
- **Aquifer** - rock / soil that stores and **yields water in useful quantity** (sand, gravel, jointed basalt, weathered granite, cavernous limestone).

- **Aquiclude** - stores water but does not transmit (clay); **Aquitard** - poorly permeable (silty clay); **Aquifuge** - neither stores nor transmits (massive granite).
- **Unconfined (water-table) aquifer** - top boundary is the water table, open to the atmosphere. **Confined (artesian) aquifer** - sandwiched between two impermeable layers; water under **pressure**.
- **Perched water table** - local water table above the main one, held on a lens of clay.

1.2 Water Table and Zones

- **Water table** = the **upper surface of the zone of saturation** at which water pressure equals atmospheric pressure. It is **not horizontal**: it **follows the topography** in a subdued form (higher under hills, lower in valleys).
- **Zone of aeration (vadose zone)** - above the water table: soil water belt, intermediate belt, capillary fringe; pores contain **air + water**.
- **Zone of saturation (phreatic)** - below the water table; all pores filled with water.
- **Fluctuation:** rises in monsoon (recharge), falls in summer (discharge, pumping, evaporation). Depth measured in open wells / **piezometers**.
- **Gaining stream** - river fed by ground water (water table slopes to river); **Losing stream** - river recharges the water table.

1.3 Cone of Depression

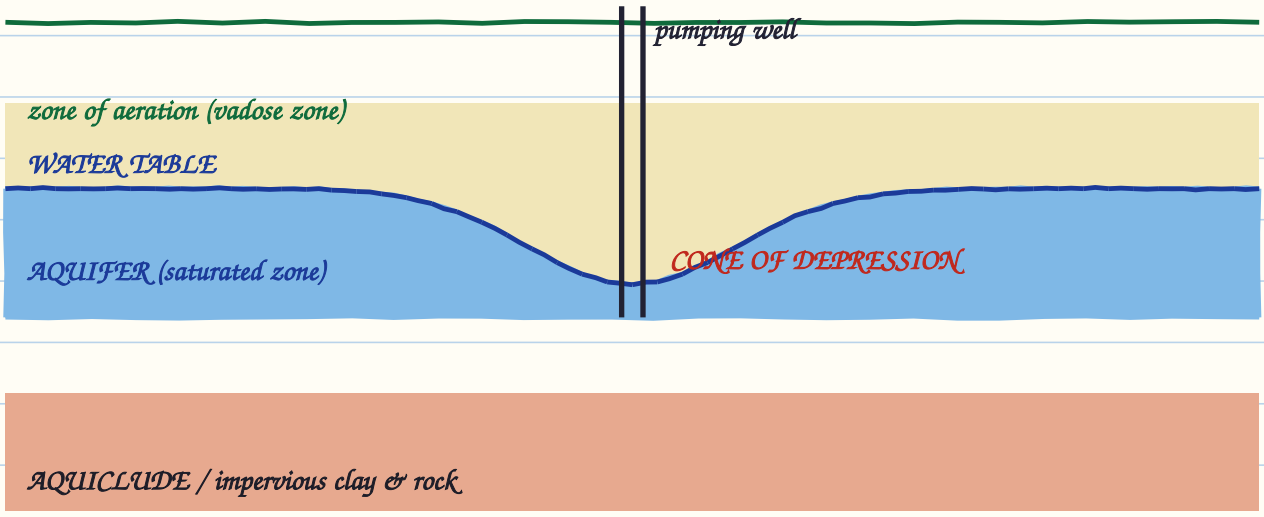


Fig 1: Water table and cone of depression around a pumping well

- When a well is **pumped**, the water level in the well drops and a **cone-shaped depression** forms in the water table around the well - the **cone of depression**.
- **Drawdown (s)** = lowering of water level (static level - pumping level). **Radius of**

influence (R) = distance from the well where drawdown becomes zero.

- Size depends on: **pumping rate**, aquifer **permeability / transmissivity**, thickness, pumping time. Fine-grained aquifer -> **steep, narrow cone**; coarse aquifer -> **gentle, wide cone**.
- **Interference:** cones of nearby wells **overlap**, increasing drawdown and well failure - wells must be **spaced** adequately.
- Over-pumping -> **declining water table, dry wells, land subsidence, sea-water intrusion** in coastal areas (e.g., Krishna / Godavari deltas, A.P. coast).
- Discharge from a well (steady flow): $Q = \pi K (H^2 - h^2) / \ln (R / r)$ for unconfined aquifer (Dupuit's formula).

Remember (diagram!)

- Draw: ground surface, water table, pumping well, drawdown curve, cone of depression, radius of influence, aquifer and aquiclude.

1.4 Geological Controls of Ground Water Movement

Movement and storage depend on the **geological setting**:

- **Lithology (rock type):**
  - **Sedimentary:** sandstone, gravel, alluvium - good aquifers; shale, clay - barriers; **limestone** - karst conduits.
  - **Igneous & metamorphic:** hard crystalline rocks have **negligible primary porosity** - water in **joints, fractures, weathered zone**. **Vesicular basalt, jointed basalt (Deccan)** yield water.
- **Porosity, permeability, grain size & sorting:** well-sorted coarse material = high yield; poorly sorted = low.
- **Structure:**
  - **Folds:** synclines collect water (artesian conditions); anticlines drain.
  - **Faults:** may act as **conduits** (breccia) or **barriers** (gouge); springs occur along faults. **Joints** give secondary porosity.
  - **Dykes** (impermeable) divide aquifers and form **dyke-ponded water**; **unconformities** act as flow paths.
  - **Dip of beds:** water moves along the dip.
- **Topography & drainage:** gentle slopes and valleys favour recharge; steep slopes give

run-off.

- **Weathering:** weathered zone (up to 20-30 m) of granite / gneiss is the **shallow aquifer** in Rayalaseema.
- **Climate & vegetation** control recharge.
- **Springs:** natural outflow where the water table meets surface - **contact springs, fault springs, depression springs, artesian springs.**

1.5 Ground Water Exploration Techniques

1. **Geological methods:** study of **rock types, structures, lineaments, geomorphology, springs, vegetation; geological map** and well inventory. Best sites: **valley fills, alluvium, intersection of fractures (lineaments), contact of dyke and country rock.**
2. **Remote sensing & GIS:** satellite images and aerial photos - **lineaments, drainage pattern, land use**, geomorphic units; used by NRSC for ground-water prospect maps.
3. **Geophysical methods (surface):**
   - **Electrical resistivity (VES)** - most widely used: low resistivity = clay or saturated zone / saline; high = dry rock. **Wenner & Schlumberger** arrays.
   - **Seismic refraction** - depth to bed rock / water table; **Gravity / magnetic** - buried valleys, dykes; **Ground-penetrating radar**, EM methods.
4. **Test drilling and well logging: exploratory bore-holes, electrical logs** (resistivity, self-potential, gamma log), **pumping tests** to find yield and transmissivity.
5. **Hydro-geochemical & isotope methods:** water chemistry (TDS, EC, ions), tracers for age and flow direction.
6. **Water divining / dowsing** - non-scientific.

Practical tip

- In granite-gneiss terrain (Anantapur-Kurnool belt) bore wells succeed along **fractures and lineaments**, not on massive rock; ideal depth 60-100 m (weathered zone + fractured zone).

2. Earthquakes

An **earthquake** is **sudden shaking / vibration of the earth's crust** due to release of strain energy stored in rocks, which travels as **seismic waves**.

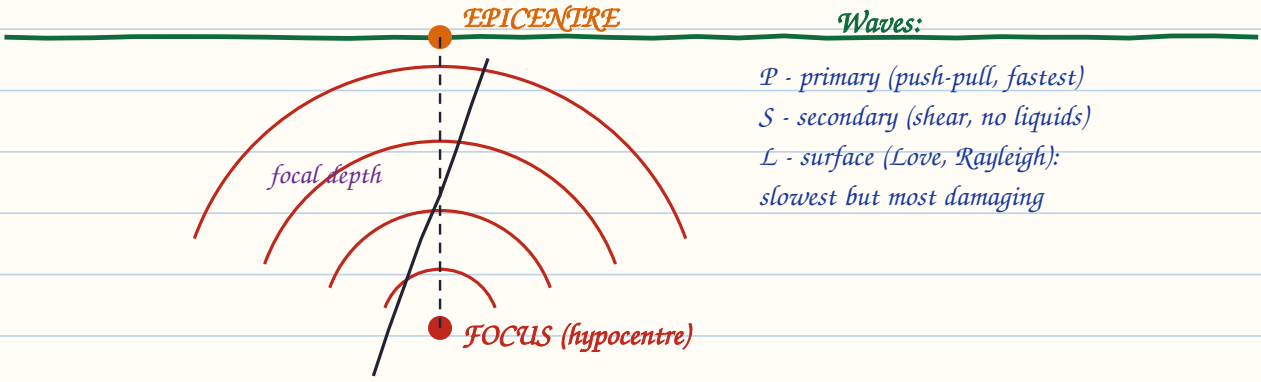


Fig 2: Focus, epicentre and seismic waves

2.1 Terminology

- **Focus (hypocentre)** - point **inside the earth** where rupture starts; **Epicentre** - point on the surface **directly above** the focus.
- **Focal depth:** shallow (< 70 km), intermediate (70-300), deep (300-700 km). Shallow quakes are most destructive.
- **Seismograph** records waves as **seismogram**; **Seismology** = science of earthquakes; **Isoseismal lines** - join places of equal intensity; **Aftershocks / foreshocks.**
- **Body waves: P-waves** (primary, compressional, **fastest**, pass through solids, liquids, gases); **S-waves** (secondary, shear, **only solids**, 1.7 × slower).
- **Surface waves: Love** (horizontal shear) and **Rayleigh** (rolling) - slowest, **cause maximum damage**.
- **S-P time lag** gives distance of epicentre; three stations -> **triangulation** locates epicentre.

2.2 Classification

- **By cause: Tectonic** (fault movement - most common, 90%), **volcanic**, **collapse** (caves, mines), **explosion** (nuclear, blasts), **reservoir-induced** (Koyna 1967, M 6.5).
- **By focal depth:** shallow, intermediate, deep.
- **By magnitude:** micro (< 3), minor (3-3.9), light (4-4.9), moderate (5-5.9), strong (6-6.9), major (7-7.9), **great (≥ 8)**.

2.3 Causes of Earthquakes

- **Plate tectonics** - movement at **convergent, divergent and transform boundaries**; **elastic rebound theory** (H. F. Reid) - rock strains, then snaps along a fault releasing energy.
- Volcanic eruptions, **isostatic adjustment**, landslides, **reservoir impoundment**, mining,

nuclear blasts, **fluid injection**.

2.4 Effects of Earthquakes

- **Primary:** ground shaking, **surface rupture**, **tilting & subsidence**, **liquefaction** (saturated sand behaves as liquid), **landslides**.
- **Secondary:** **collapse of buildings, bridges, dams**, fires, floods, **tsunami** (sea-floor quakes - Indian Ocean 2004), **damage to pipelines**, loss of life, disease.
- Famous Indian quakes: **Latur 1993 (M 6.2)**, **Bhuj 2001 (M 7.7)**, **Uttarkashi 1991**, **Kashmir 2005**, **Sikkim 2011**, Assam 1950.

2.5 Shield Areas and Seismic Belts

- **Shield** = ancient, **stable crust** of Precambrian crystalline rocks; **low seismicity**. **Peninsular India** (Dharwar, Bastar, Singhbhum cratons) is a stable shield - **but not quake-free** (Latur, Koyna, Jabalpur, Bhadrachalam).
- **Seismic belts (zones):** (i) **Circum-Pacific belt** ("Ring of Fire", ~ 80% of energy), (ii) **Alpine-Himalayan (Mediterranean-Trans-Asiatic) belt** ~ 15%, (iii) **Mid-Atlantic ridge / mid-oceanic ridges**.
- **Seismic zones of India (IS 1893:2016): Zone II** (low), **III** (moderate), **IV** (severe), **V** (very severe: Himalaya, NE India, Kutch, Andaman). Zone I does not exist now.
  - **Andhra Pradesh** lies mostly in **Zone II and III** (Kurnool, Anantapur - Zone II/III).

2.6 Magnitude and Intensity

Feature	Richter scale (magnitude)	Mercalli scale (intensity)
Measures	**Energy released** at the focus	**Effects / damage** at a place
Type	Quantitative, logarithmic, instrument-based	Qualitative, I - XII, observation
Variation	One value for a quake	Varies with distance and ground
Meaning	Each **+1 = 10 times amplitude, ≈ 32 times energy**	I = not felt ... **XII = total destruction**

- **Richter (1935):** $M = \log A$ + correction; $M < 2$ not felt; 5 - damage to weak buildings; 7 - serious damage; > 8 - great. Now **moment magnitude (Mw)** is preferred for large quakes.
- **Modified Mercalli (MMI):** I (instrumental) - III (feel indoors) - V (waking) - VII

(damage to poor buildings) - IX (heavy damage) - XII (total).

2.7 Precautions for Building Construction in Seismic Areas

- **Site selection:** avoid **active faults, loose sand (liquefiable), soft clay, steep slopes, filled ground**; prefer **rock or firm soil**.
- **Foundation:** uniform and on **same type of soil**; **raft / strip with tie-beams**; deeper than weathered zone; avoid partial pile / footing mix.
- **Configuration: simple, symmetrical, regular** plan; avoid soft storeys, projections, re-entrant corners, heavy top, large openings; **adequate separation (expansion) joints** between buildings.
- **Materials & detailing:** use **light weight** materials; **ductile detailing (IS 13920)**; strong column - weak beam; **shear walls**, bracing; **bands** (plinth, lintel, roof) and vertical bars in masonry (**IS 4326**).
- **Design:** use **IS 1893** zone factors (Z); **base isolators, dampers.**
- **Special structures:** dams - **flexible earth-fill or gravity dams on rock**, seismic coefficient method; tall chimneys, bridges with **seismic arrestors**.
- Public: **early warning, drills, retrofitting** of old buildings.

3. Landslides

Landslide = the **downslope movement of a mass of rock, debris or earth under the influence of gravity. Mass wasting** is the general term (includes creep, flows, falls).

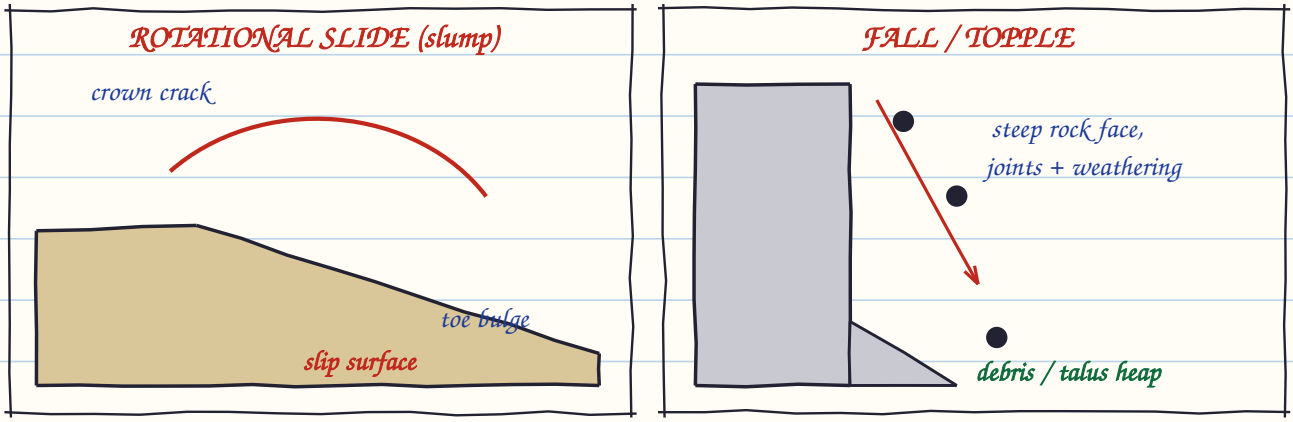


Fig 3: Rotational slide and rock fall

3.1 Classification of Landslides (Varnes / IS 14680)

Type	Description
Falls	Free fall / bouncing of rock and debris from **steep cliffs**; sudden; rock fall, debris fall
Topples	**Forward rotation** of rock columns about a pivot point
Slides	Movement on a **surface of rupture. Rotational (slump)** - curved surface, in homogeneous soil; **translational / planar** - along bedding, joint or fault plane; **wedge** slide - along two intersecting planes
Flows	Material behaves like a fluid - **debris flow, mud flow, earth flow, avalanche**; with heavy rain
Creep	Very **slow** continuous movement of soil, shown by tilted poles, curved tree trunks
Lateral spreading	Extension of soil mass over soft layer - in earthquakes (liquefaction)
Complex	Combination of two or more types

- Also classified by **speed** (very slow < 16 mm/year to extremely rapid > 5 m/s), **material** (rock, debris, earth) and **water content**.

3.2 Causes of Landslides

- **Natural causes:**
  - **Gravity** acting on **steep slopes**; **Heavy rainfall / snow melt** - raises pore pressure, adds weight, reduces shear strength (monsoon slides in Western Ghats, Himalaya).
  - **Geology: weak rocks** (shale, clay, schist, phyllite), **weathered rock, jointed and sheared rock, bedding / joints dipping out of the slope** (daylighting), presence of faults, alternate hard-soft beds.
  - **Earthquakes** and volcanic shocks; **erosion of toe** by river / sea waves; **ground water** rise.
- **Human (anthropogenic) causes:**
  - **Excavation / cutting at the toe** of slopes for roads and buildings (over-steepening); **loading at the top** (buildings, dumps); **deforestation**;
  - **Blasting, mining, quarrying**; **leakage from canals, pipes, reservoir drawdown**; **improper drainage and irrigation.**

3.3 Effects of Landslides

- Loss of life and property; **blocking of roads, railways, rivers** (landslide dams -> floods when breached).
- **Damage to dams, reservoirs, canals, power lines** - sudden waves (Vajont).
- Loss of **agricultural land**, forests, soil fertility; **siltation** of reservoirs; **economic loss**, disruption of transport, tourism.
- Examples in India: **Malpa (1998), Kedarnath (2013), Malin (Pune, 2014), Wayanad (2024), Himalayan highways yearly**.

3.4 Preventive and Control Measures

- **Hazard mapping & zonation** (BIS, GSI) - avoid construction in high-hazard zones; **monitoring** (inclinometers, extensometers, rain gauge, **early warning**).
- **Surface and sub-surface drainage: catch-water drains**, lined side ditches, **horizontal drains (weep holes)**, drainage galleries - the **most effective** measure.
- **Slope modification: reduce slope angle**, **benching / terracing**, remove top load (**unloading**), **fill at the toe (counterweight berm)**.
- **Retaining structures: retaining walls, gabion walls, breast walls, crib walls, soil nailing, rock bolts and anchors, shotcrete, wire-mesh**, piles.
- **Bio-engineering: afforestation**, grass turfing, vetiver, **jute / coir geotextile**.
- **Rock fall control: catch fences, rock sheds, trenches**, scaling of loose rocks.
- **Good practice:** do **not cut slopes at too steep angles**; avoid blasting; **plan roads along stable contours**; **control irrigation and leakage**; proper **spoil disposal**.

Remember (ABC of slope stability)

- A - Angle of slope B - Bedding / joint orientation C - Climate / water -> Reduce driving force, increase resisting force (drain, support, vegetate).

4. Geophysics

Geophysics applies **physics (gravity, magnetism, electricity, elasticity, radioactivity)** to study the **earth's subsurface** from surface (or air / bore-hole) measurements, without direct excavation. **Exploration geophysics** looks for physical-property **contrasts**: density, magnetic susceptibility, resistivity, wave velocity.

4.1 Importance of Geophysical Methods

- **Rapid, non-destructive and cheap** compared with drilling; covers large areas.
- Gives **depth to bed rock**, thickness of overburden, **water table**, fracture / fault zones, **cavities**, buried channels - for **dams, tunnels, foundations, roads**.
- Used in ground water, mineral, oil and gas exploration, archaeology and pollution mapping.
- Limitations: **indirect** (needs interpretation, ambiguity), calibrated with **bore holes**.

4.2 Classification

Method	Property measured	Instrument	Engineering use
Gravity	Variation in **density**, g	Gravimeter	Large structures, caverns, salt domes, ore bodies
Magnetic	Magnetic susceptibility	Magnetometer	Dykes, buried faults, iron ores
Electrical	**Resistivity / conductivity**	Resistivity meter (SP, IP)	Ground water, bed rock depth, cavities
Seismic	**Wave velocity**	Geophones, seismograph	Rock quality, layers, depth to rock
Radiometric	Natural **gamma radiation**	Scintillation counter, Geiger	Uranium, thorium, clay layers, mapping

- **By source: natural source** (gravity, magnetic, SP) and **artificial (active)** (resistivity, seismic). **By location:** ground, airborne, marine, bore-hole.

4.3 Principles of Methods

(a) Gravity Method

- Based on **Newton's law**: $g = G\,M / R^2$. **Denser bodies** (ore, rock) give **positive gravity anomalies**; voids / light rock (salt, caves) give **negative** anomalies.
- Measurement: **gravimeter** at stations; corrections: **latitude, elevation (free-air, Bouguer), terrain, tidal drift**. Units: **mGal**.

(b) Magnetic Method

- Based on **differences in magnetic susceptibility** causing local variations in earth's magnetic field. Magnetite, basic rocks - **high anomalies**; sedimentary - weak.
- **Magnetometer** (proton precession, fluxgate), **unit = nT (nanotesla)**. Corrections:

diurnal variation, IGRF.

(c) Electrical Methods

- **Electrical resistivity method:** current *I* is passed through the ground by two current electrodes, **potential difference *V*** is measured by two potential electrodes. **Apparent resistivity** rho-a = K (V / I); for **Wenner array** rho = 2pi a (V/I).

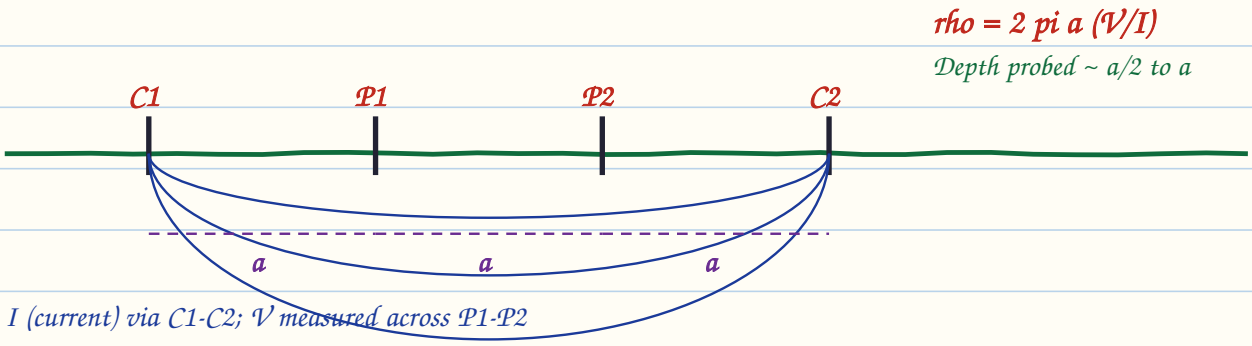


Fig 4: Wenner electrode array

  - **Vertical Electrical Sounding (VES)** - electrodes spacing increased about a centre -> **depth information**; **Profiling** - fixed spacing moved along line -> **lateral changes**.
  - Typical resistivity (Ω·m): **clay 1-20, wet sand 50-200, saline water < 5, weathered rock 20-100, granite 1000+, dry limestone 500-5000.**
- **Self-potential (SP), Induced polarization (IP), EM methods** also used.

(d) Seismic Methods

- Based on **propagation of elastic waves**: artificial source (hammer, explosion), receivers (**geophones**). Velocity increases with **density and rigidity**: soil 200-800 m/s, weathered rock 800-2000, **hard rock 3000-6000 m/s**, water ≈ 1500.
- **Seismic refraction method** - waves refract at interface where **V2 > V1** (critical refraction); gives **depth and velocity** of layers.

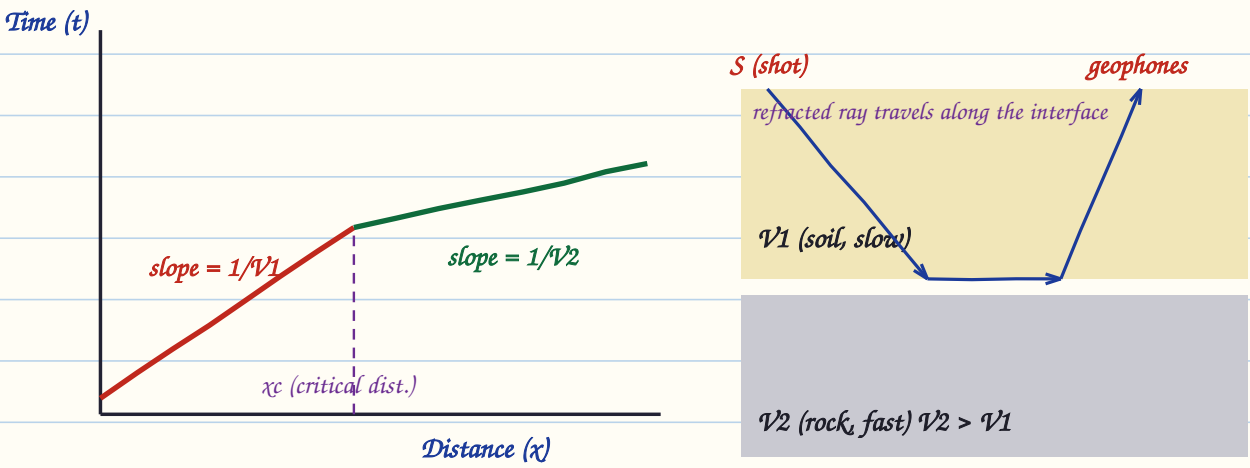


Fig 5: Seismic refraction - time-distance graph

- **Depth of first layer:** $h = (xc / 2) \times \sqrt{[(V2 - V1)/(V2 + V1)]}$ (xc = critical distance) or $h = (ti \times V1\ V2) / (2\sqrt{(V2^2 - V1^2)})$.
- Limitations: **fails if a low-velocity layer lies under a high-velocity layer (blind zone).**

- **Seismic reflection** - echo from boundaries, deep structure (oil exploration).

(e) Radiometric Method

- Measures **natural gamma radiation** from radioactive minerals (**U, Th, K-40**) with **scintillometer / Geiger-Müller counter**; also **gamma-gamma density logging**; used for **uranium exploration** and mapping lithology (e.g., granite, shale).

4.4 Electrical Resistivity vs Seismic Refraction (Quick Comparison)

Aspect	Resistivity	Seismic refraction
Property	Ground resistivity (Ω·m)	Wave velocity (m/s)
Best for	**Ground water, clay layers, saline zones**	**Bed rock depth, rock rippability**
Limitations	Ambiguity from equivalence; metal pipes disturb	Needs velocity increasing with depth; noise
Cost	Low	Moderate

5. Engineering Properties of Rocks

- **Density & specific gravity:** granite 2.6-2.7, basalt 2.8-3.0; **higher density = higher strength**, durability.

- **Porosity & water absorption:** low in igneous rocks (< 1%); high in sandstone and limestone. Good building stone absorption **< 5%** (IS: **< 0.6% for heavy** structures); high absorption -> **frost damage**.
- **Permeability:** primary (pores) and **secondary (joints)** - leakage in reservoirs.
- **Strength: compressive strength** (UCS) is main parameter; tensile strength is only ~ **1/10** of compressive; **shear strength**. **Rock hardness**.
- **Elasticity:** Young's modulus 20-100 GPa for hard rock, Poisson's ratio 0.1-0.3; **elastic, brittle**.
- **Durability / weathering resistance:** resistance to wetting-drying, acids, frost; **slake durability index** - shale low.
- **Abrasion & toughness (Los Angeles abrasion, impact value)** - road aggregate (limit ≈ 30-35%).
- **Anisotropy:** schist, slate, shale behave differently along and across planes.
- **RQD (Rock Quality Designation)** = (sum of core pieces > 10 cm / total core length) × 100.

RQD (%)	Rock quality
< 25	Very poor
25-50	Poor
50-75	Fair
75-90	Good
90-100	Excellent

- **Other rock mass classifications: RMR (Bieniawski), Q-system (Barton)** for tunnels; **Geological Strength Index (GSI)**.

Exam points

- Q: Cone of depression, Richter scale vs Mercalli, landslide control, resistivity method, seismic refraction - each 5-10 marks. Draw sketch + formula + engineering use.

5.1 Short Revision Questions

1. **Define water table.** Upper surface of zone of saturation.
2. **What is an aquifer / aquiclude?** Water-yielding / water-storing but non-transmitting rock.
3. **Define focus and epicentre.** Focus - origin below; epicentre - surface point above it.

4. **Which waves cause maximum damage?** Surface waves (Love, Rayleigh).
5. **Seismic zones of India?** II, III, IV, V.
6. **Name two landslide causes due to man.** Slope cutting, blasting, deforestation, poor drainage.
7. **Principle of resistivity method?** Different rocks have different electrical resistivity; rho = 2pia V/I (Wenner).
8. **Seismic refraction condition?** V2 > V1 (velocity must increase with depth).

ENGINEERING GEOLOGY

UNIT - V : DAMS, RESERVOIRS, TUNNELS

Geology of Dams, Reservoirs and Tunnels

Syllabus (as given)

- Types and purpose of dams; geological considerations in the selection of a dam site; geological considerations for successful construction of reservoirs, life of reservoirs; purpose of tunnelling, effects, lining of tunnels; influence of geology for successful tunnelling.

1. Dams - Types and Purpose

A **dam** is a **barrier constructed across a river or valley to store water**, forming a **reservoir**. India has > 5000 large dams (3rd in the world); e.g., **Nagarjuna Sagar, Srisailam, Tungabhadra, Somasila, Bhakra, Tehri, Hirakud**.

1.1 Purposes of Dams

- **Irrigation** (storage for canals), **drinking and industrial water supply**.
- **Hydro-electric power** generation.
- **Flood control / moderation** and **river regulation**.
- **Navigation, fisheries, recreation, tourism**, ground-water recharge, **silt control**.
- **Multi-purpose dams** serve more than one aim (Nagarjuna Sagar: irrigation + power + flood control).

1.2 Types of Dams

- **Based on material: masonry / concrete** (gravity, arch, buttress), **earth-fill, rock-fill**, steel, timber.
- **Based on structural action:**

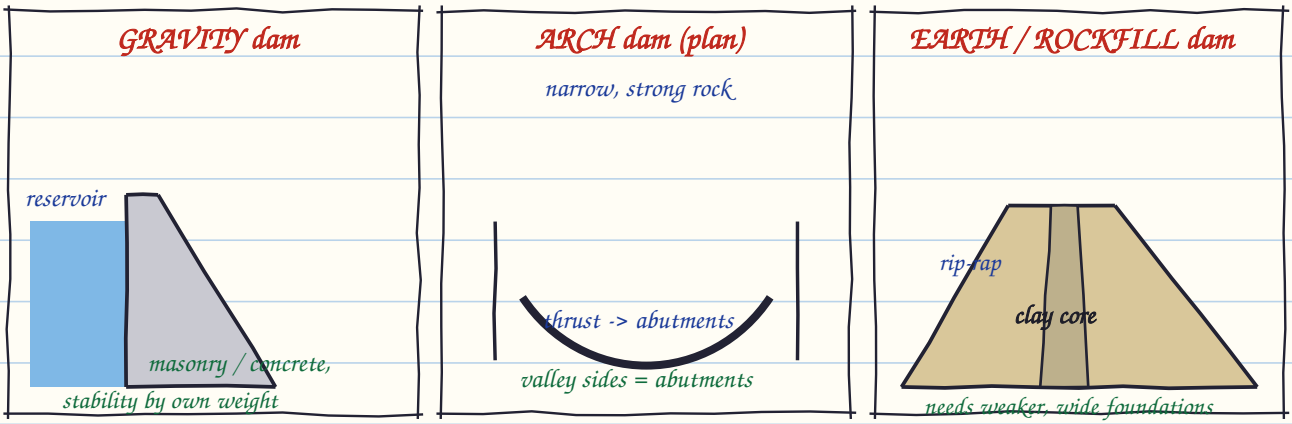


Fig 1: Gravity, arch and earth / rock-fill dams

Type	Features and suitable geology
Gravity dam	**Resists water thrust by its own weight**; masonry / concrete; triangular section; needs **strong, sound rock foundation** (granite, gneiss, quartzite, massive basalt) - very high bearing capacity. Eg: Nagarjuna Sagar, Bhakra
Arch dam	**Curved in plan**, transfers thrust to **abutments**; thin section, less concrete; needs **narrow gorge with strong, sound, unjointed rock walls**. Eg: Idukki (Kerala), Vajont
Buttress dam	Water face supported by triangular buttresses; needs **good rock at buttress points**; less concrete than gravity
Earth-fill (embankment) dam	**Compacted earth** with **impervious core** and shell; **flexible**; can be built on **weaker, deformable foundations (alluvium, soil)**; suitable in **wide valleys** and seismic zones. Eg: Somasila, Tungabhadra (partly)
Rock-fill dam	Rock dumped / compacted with an **impervious membrane or core**; use where good rock is available; Eg: Tehri (earth + rock fill)
Composite dam	Combination - masonry / concrete section + earth embankment (Nagarjuna Sagar, Tungabhadra)

- **Other classification:** by **hydraulic design** (overflow / non-overflow), by **height** (low < 30 m, medium 30-90 m, high > 90 m).

Foundation rule

- Concrete gravity / arch dams -> hard, strong rock. Earth / rock-fill -> any competent foundation incl. soil. The **higher the dam, the better the foundation must be**.

2. Geological Considerations in Selection of a Dam Site

A good dam site must satisfy **geological, topographical, hydrological and economic** conditions. The geologist studies **rock type, structure, strength, permeability and weathering** of the foundation and abutments.

2.1 Topographic Conditions

- **Narrow valley (gorge)** with **wide reservoir basin** upstream (economical length, large storage).
- Strong, **sound rock at shallow depth** in river bed and abutments; **good approach** and availability of **construction materials** (aggregate, clay, sand) nearby.

2.2 Geological Considerations

1. **Rock type & strength**
   - Foundation rock must have **adequate compressive strength, low compressibility** and bearing capacity > dam load: **granite, gneiss, quartzite, dolerite, massive basalt** are excellent.
   - **Poor rocks:** shale, schist, slate, phyllite, weathered / soft sandstone, friable rocks - **slake, swell, slide**; **limestone** - cavities; **gypsum, rock salt** - soluble, unsuitable.
2. **Geological structures**
   - **Attitude of beds (strike & dip):** beds **dipping upstream** are favourable (water pressure and thrust press the dam into rock; leakage resisted); **dip downstream** -> **seepage, uplift, sliding** risk; **horizontal beds** - safe if strong; **steeply dipping beds with strike parallel to the river** - leakage.
   - **Folds:** avoid dam in **anticline crest**; prefer **syncline limb** with moderate dip; unfavourable fold may cause **leakage and instability**.
   - **Faults: avoid active / major faults** - weak crushed zone, **seepage, differential settlement, seismic displacement**. Minor faults can be treated (concrete plug, grouting).

- **Joints & fractures:** closely spaced, open, persistent joints cause leakage and low strength - treat by curtain grouting, consolidation grouting; joint sets dipping towards valley reduce abutment stability.
- **Unconformities, dykes, shear zones:** weak planes; dykes may stop or allow seepage - study.

3. **Weathering:** deep weathered rock - excavation to fresh rock (increases cost); thickness of overburden / alluvium must be small; bore holes.
4. **Permeability & ground water:** high permeability foundation -> seepage, piping, uplift pressure; ground-water level and water tightness of reservoir rim. Karst limestone areas often leak.
5. **Abutment stability:** slopes should be stable with no landslide mass (Vajont - study of reservoir rim); check dip of beds towards valley.
6. **Seismicity:** area should be tectonically stable; design for seismic coefficient; avoid active faults and reservoir-induced seismicity zones.
7. **Construction materials:** availability of suitable aggregates, sand, clay, rock nearby; quarry geology.
8. **Reservoir basin:** water-tight, with no leakage and limited siltation (see next section).

Rock	For concrete dam	Remarks
Granite / Gneiss	Excellent	Sound; grout joints
Basalt (massive)	Very good	Check vesicular / jointed layers
Quartzite / Dolerite	Excellent	Hard; may be highly jointed
Sandstone	Good if hard	Check cement; bedding planes
Limestone	Doubtful	Cavities, solution channels
Shale / Schist / Slate	Poor	Sliding, swelling; weak planes

2.3 Foundation Treatment

- **Excavation** to sound rock; **cut-off trench** (earth dams); **grout curtain** (cement grouting in a line of holes) to **reduce seepage**; **consolidation grouting** to strengthen jointed rock; **drainage holes** to reduce **uplift**; **dental concrete** in cavities; **shear keys**, rock bolts / anchors.
- **Dam failures** studied earlier (St. Francis, Malpasset, Teton, Vajont) show how neglect of geology leads to disaster.

2.4 Geological Causes of Dam Failures

- **Foundation failure:** shear along weak beds (shale / clay), **excess settlement** on soft strata, **sliding** on bedding or joint planes dipping downstream.
- **Seepage and piping:** flow through **joints, faults, solution cavities, permeable layers** -> removal of fines -> tunnel-like channel -> breach (Teton).
- **Uplift pressure** under the dam raised by water in joints (Malpasset).
- **Abutment failure** - wedge failure along intersecting joints / faults; **slope failure** on reservoir rim.
- **Earthquake shaking / fault movement** and **overtopping** due to floods beyond spillway capacity.
- **Slaking / softening** of rock on contact with water (St. Francis conglomerate).

2.5 Indian Dam Examples (geology)

Dam / River	Type	Geological setting
Nagarjuna Sagar (Krishna)	Masonry gravity + earth flanks	Hard Precambrian rocks of Nallamala region
Srisailam (Krishna)	Concrete gravity	Narrow gorge in quartzite / crystalline rocks
Idukki (Periyar)	Double-curved **arch**	Narrow gorge in **massive charnockite**
Tehri (Bhagirathi)	Earth + rock-fill (260 m)	**Seismic zone IV**; phyllite and quartzite of lesser Himalaya
Somasila (Penna)	Earthen	Wide valley, Cuddapah / granite-gneiss terrain
Koyna (Koyna)	Rubble-concrete gravity	**Deccan basalt**; RIS 1967

3. Reservoirs

A **reservoir** is an **artificial lake** formed by storing water behind a dam. **Purposes:** water storage (irrigation, supply), hydropower, flood regulation, recreation.

3.1 Geological Considerations for Successful Reservoirs

- **Water-tightness (no leakage):** reservoir basin should be formed in **impervious rocks** (shale, clay, massive granite, gneiss) or by **impermeable beds**. **Avoid** karst limestone,

highly jointed basalt, fractured quartzite, coarse alluvium on the rim.

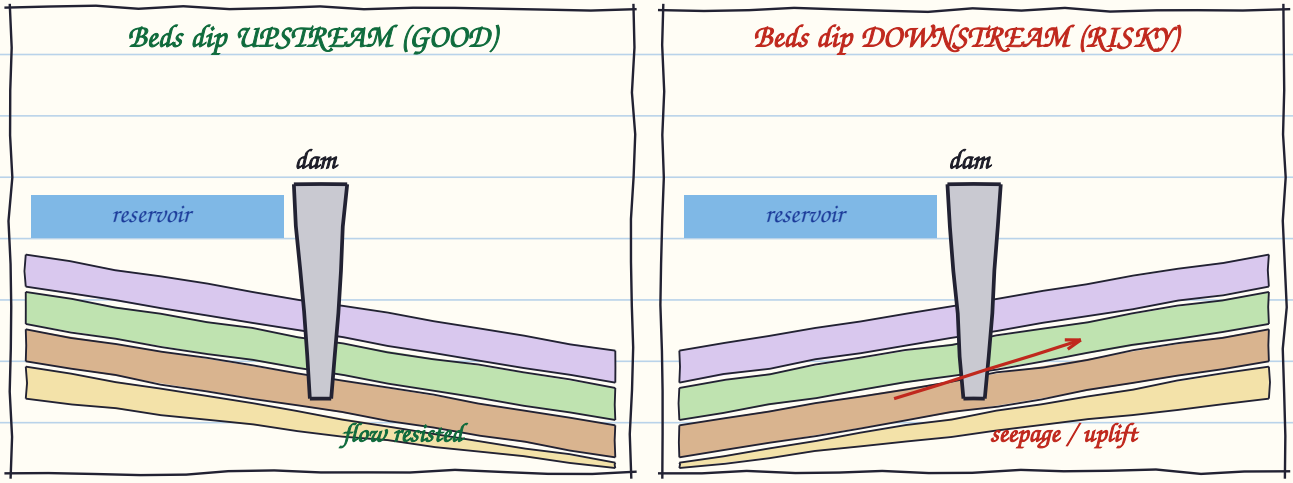


Fig 2: Effect of dip of beds on seepage under the dam

- **Structure of rocks:**
  - **Dip of strata:** beds dipping **upstream / towards the reservoir** and **water-bearing strata dipping away** -> **leakage possible**; beds dipping **downstream** and sloping towards valley -> **water escapes** (risk) if permeable; **impervious beds dipping upstream** are best.
  - **Faults / joints** crossing the rim or the valley can act as leakage channels; **dykes** may be barriers.
  - **Anticline** - leakage through the limbs; **syncline** - water-tight, favourable if impermeable beds.
- **Water table:** if the **ground water table lies above the reservoir level** around the rim -> **no leakage** (water table slopes towards the reservoir). If the water table is **below reservoir level and slopes away** (steep downstream) -> **leakage** (e.g., in basins with deep water table).
- **Stability of reservoir slopes (rim):** unstable rim -> **landslides into the reservoir** (Vajont) -> waves and loss of capacity; study **slope, weathering, strata, ground-water conditions**.
- **Seismicity:** avoid **active fault zones**; **reservoir-induced seismicity (RIS)** (Koyna, Kariba, Hoover) after impounding of **deep reservoirs (> 100 m)** on pre-stressed faults.
- **Siltation:** rock type and land use in **catchment** control silt load; **soft rocks, steep slopes, deforestation** produce silt -> **reduces storage** capacity.
- **Water quality:** soluble rocks (**gypsum, rock salt**) may spoil water chemistry; dissolved

minerals.

- **Large storage capacity:** broad valley behind a narrow dam gorge, with sealed rim.
- **Submergence:** minimise loss of land, forests, mineral resources, historic sites; **rehabilitation**.

3.2 Treatment of Leakage

- **Grouting (curtain)** and **blanketing** (clay blanket on the upstream floor), **cut-off walls**, **sealing of springs**, **concrete lining**, plugging of sinkholes; careful **site selection**.

3.3 Life of Reservoirs

- **Storage capacity** declines because **sediment (silt) enters and settles** in the reservoir - **reservoir sedimentation**. Life = time to fill the **dead storage** (and then live storage) with silt.
- Sedimentation rate depends on: **catchment area, slope, rock type, soil erosion, rainfall, vegetation**, land use, size of reservoir, **trap efficiency** (fine particles escape).
- **Life (years) ≈ Total storage capacity / Average annual silt inflow**. Typical **design life 50-100 years**; Indian reservoirs lose ~ **0.5-1% capacity per year**.
- **Zones of reservoir: dead storage** (below minimum draw-down level), **live storage** (useful), **flood storage** (surcharge).
- **Control of sedimentation: catchment treatment** (afforestation, check dams, contour bunding, terracing), **silt traps / sediment ponds**, **dredging**, **sluicing and flushing**, **bypass of silty floods**, **upstream sediment detention dams**, **optimum operation**.

Examples

- Nagarjuna Sagar (Krishna River, A.P./Telangana) - 124 m high masonry gravity dam founded on hard **Precambrian rocks**; Srisailam - concrete gravity dam in a narrow **gorge through the Nallamala hills**; Tungabhadra - heavy silting has reduced its storage capacity.

3.4 Reservoir-Induced Seismicity (RIS)

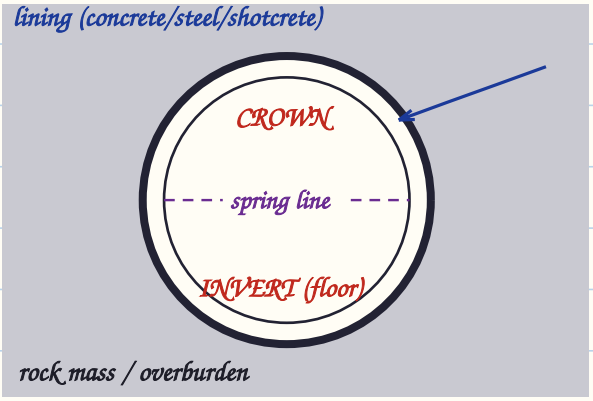
- **Earthquakes triggered by impounding** of a reservoir - pore-pressure rise along existing faults + extra load of water.
- Known cases: **Koyna (India, 1967, M 6.5)**, Kariba (Zambia, 1963), Hoover (USA), Aswan, Hsinfengkiang (China).
- Favourable conditions: **depth of water > 100 m**, large volume, **pre-existing fractures**,

rapid filling.

- Precautions: **seismic study of region**, **install seismographs**, **fill slowly in stages**, flexible dam design.

4. Tunnels

A **tunnel** is an **underground passage (horizontal or nearly so) excavated through rock or soil**, **without removing the overburden**, with a cross-section generally closed.



- Circular - soft rock, water pressure
- Horseshoe - roads / railways
- D-shaped / arched - hard rock
- Rectangular - cut & cover, metro

Shape depends on rock quality, use and construction method

Fig 3: Tunnel cross-section and shapes

4.1 Purpose of Tunnelling

- **Transportation: railway, highway** tunnels through mountains (Konkan Railway, Jawahar tunnel, Rohtang Atal tunnel), **metro** (urban).
- **Water conveyance: irrigation, hydro-power (head race / tail race) tunnels**, **water supply** and **sewage** tunnels (e.g., the **SLBC and Veligonda project tunnels** in the Krishna basin).
- **Mining** (adits, drifts), **military and defence**, **storage** (oil, gas, nuclear waste), **utility tunnels** (cables, pipes), **pedestrian subways**.
- **Advantages:** shorter route, avoid steep gradients, protect from weather / landslide, preserve surface land, save time and fuel.

4.2 Effects of Tunnelling (on rock and environment)

- **Stress redistribution** around the opening -> **roof fall, rock burst, squeezing / swelling** of rock, **over-break**.
- **Ground water: inflow** (flooding), **lowering of the water table** (drying of wells and springs), **subsidence** of surface (urban tunnels), **blasting vibration**.

- **Gases:** methane, H_2S, CO_2 in coal / sedimentary rocks; **heat** in deep tunnels (geothermal gradient, 1° C per 30 m).
- **Environmental:** disposal of **muck**, dust, noise, vibration, ecological impacts.

4.3 Lining of Tunnels

Lining = a **supporting layer** put on the tunnel surface to **prevent collapse, resist rock and water pressure, reduce friction (water tunnels) and give a smooth, durable surface.**

- **Purposes:** support the **roof & walls**, prevent **fall of loose rock**, exclude water (**water-tightness**), give **smooth flow** in conduits, **protect against weathering** (slaking), safe working.
- **Types and materials:**
  - **Timber / steel supports** (arches, ribs, **steel sets**) - temporary support; used in weak ground.
  - **Shotcrete (sprayed concrete)** with **wire mesh / steel fibre** and **rock bolts** - flexible lining used in **NATM** (New Austrian Tunnelling Method).
  - **Cast-in-situ (plain or reinforced) concrete** - permanent lining; **pre-cast concrete segments** - for **TBM** (tunnel boring machine) tunnels, esp. soft ground.
  - **Brick / masonry lining** - old railway tunnels; **steel liner (penstock)** - high-pressure hydropower; **cast iron segments** - metro.
- **Thickness:** from **thin (shotcrete 5-15 cm)** to **thick (concrete 30-100 cm)** depending on **ground conditions and tunnel size.**
- **Need of lining depends on geology: hard, massive rock (granite, quartzite) - unlined or partially lined; weak, jointed, weathered, water-bearing rock - full lining** essential.

Rock condition	Support needed	Example
Massive hard rock	None / spot bolts	Granite, quartzite
Fractured rock	Bolts + shotcrete + mesh	Jointed basalt
Weak / squeezing rock	Steel ribs + concrete lining	Shale, schist, fault zone
Soft ground	Segmental lining (TBM)	Alluvium, clay (metro)

4.4 Influence of Geology on Successful Tunnelling

1. **Rock type & strength: strong, massive, homogeneous** rocks (granite, gneiss, quartzite, dolerite, massive limestone, sandstone) are **ideal** - stable, self-supporting,

need little support. **Weak rocks** (shale, schist, slate, phyllite, clay, weathered rock) need heavy support.

- **Soft ground** (soil, alluvium) - **shield / TBM**; **soluble rocks** (limestone, gypsum) - caves.
- **Rock hardness affects drilling** and blasting cost; **abrasiveness** wears cutters; **rippability** (seismic velocity).

2. **Geological structures**
   - **Attitude of beds: horizontal beds** - roof stays good if strong beds above; **dipping beds** - best when the tunnel axis is **perpendicular to strike (across the strata)**; tunnel **along the strike** is risky: **roof and one wall slide** along bedding planes and **uneven pressure** develops.
   - **Folds:** in an **anticline**: tension joints, **roof falls**; in a **syncline**: good rock but **water accumulation**; best to **cross folds at right angles** and in the **limb zone** with moderate dip. Tunnel along the axis of a **plunging fold** can be good if beds are strong.
   - **Faults: major problem** - crushed rock (gouge), **water inflow**, **heavy support**, squeezing; **cross faults at high angle** and plan treatment (**probe holes, grouting, spiling**); avoid **parallel** alignment with a fault.
   - **Joints: joint orientation** affects wedge falls; **closely spaced** - over-break; tunnel **across** joint sets; **grouting** if water bearing.
   - **Unconformities & dykes:** variable ground; **dykes** may supply or block water.
3. **Ground water: inflow is the most serious problem** - flooding, **slow progress**, instability, **lining pressure**; **water-bearing joints, faults, sand layers, karst**; control by **grouting, dewatering, drainage, freezing**; tunnels below water-table need **water-tight lining**.
4. **Weathering and overburden:** deep, **weathered** zone gives poor roof; **thickness of cover** (at least **1.5-2 × tunnel diameter**) for stability; **portals** should be located on **sound rock**, away from landslide / debris slopes.
5. **Rock stress and temperature: high cover -> rock burst** (hard rock) or **squeezing** (soft rock); heat; **seismic activity**.
6. **Gases & special rocks: methane** in coal measures, **swelling clays (montmorillonite)**

soluble minerals, radon.

7. **Rock mass quality:** assessed by **RQD, RMR, Q-system** - used to decide **support**; **pre-construction exploration** (bore-holes, **seismic refraction**, resistivity) for **geological longitudinal section**.
8. **Construction method choice: drill & blast** (hard rock), **NATM, TBM** (uniform rock), **cut & cover** (shallow urban), **shield tunnelling** (soft ground).

Alignment rules (remember!)

- (i) Cross rock structures at **high angle**. (ii) Avoid **faults, shear zones & weathered ground**. (iii) Keep **portals in sound rock**. (iv) Stay above/below **aquifers** as needed. (v) Prefer **strong, homogeneous** rock.

4.5 Geological Investigation for Tunnels

- **Geological mapping** of the surface along the alignment (rock types, dip, strike, structures, springs), **aerial photo / remote sensing**.
- **Geophysical surveys** (**resistivity, seismic refraction**) for **overburden depth, faults, water**.
- **Exploratory drilling** and **core logging (RQD)**; **pilot tunnels**, **probe holes ahead of face**.
- Prepare **geological longitudinal section** and **rock mass classification** to forecast **support & cost**.
- **Case: Konkan Railway** (~ 90 tunnels) - geology-based alignment reduced collapse; **Atal Tunnel (Rohtang)** - 9 km highway tunnel built through **fractured, water-bearing Himalayan rock and the Seri Nala fault zone** with heavy support.

4.6 Rock Mass Classification for Tunnels

RMR value	Class	Support (about 10 m dia. tunnel)
81-100	I - Very good	None / spot bolts
61-80	II - Good	Local bolts, thin shotcrete
41-60	III - Fair	Systematic bolts, 5-10 cm shotcrete
21-40	IV - Poor	Bolts + 10-15 cm shotcrete + steel ribs
< 20	V - Very poor	Heavy ribs, thick lining, pre-support

- **RMR (Bieniawski)** uses 5 parameters: strength of intact rock, **RQD**, **joint spacing**, **joint condition**, **ground water**, + orientation adjustment. **Q-system (Barton)** $Q = (RQD/Jn) \times (Jr/Ja) \times (Jw/SRF)$.

4.7 Tunnel Hazards and Remedies

Hazard	Geological cause	Remedy
Roof fall / over-break	Joints, faults, weathered rock	Rock bolts, shotcrete, steel ribs
Squeezing	Soft shale / schist under high cover	Yielding supports, heavy lining
Water inflow	Fractures, karst, faults, sand	Grouting, drains, dewatering
Rock burst	Brittle rock, high stress	Destressing, support, slow advance
Gas / heat	Coal measures, deep cover	Ventilation, monitoring
Swelling	Clays (montmorillonite)	Invert lining, drainage

5. Comparison and Summary

Aspect	Dam	Reservoir	Tunnel
Key rock need	Strong, sound foundation	Impervious basin	Strong, stable, dry rock
Main structural risk	Faults, downstream dip	Leaky joints, karst	Faults, joints along axis
Water issue	Seepage and uplift	Leakage / siltation	Inflow
Treatment	Grouting, drains	Blanketing, grouting	Lining, drainage, bolts
Failure case	St. Francis, Malpasset	Vajont (rim slide)	Squeezing & collapse

5.1 Short Revision Questions

1. **Name the purposes of a dam.** Irrigation, power, water supply, flood control, navigation, recreation.
2. **Which dam needs the strongest foundation?** Arch and gravity (concrete / masonry) dams; arch dams need strong abutments.
3. **Why is upstream dip preferred?** Water pressure pushes beds into the rock, seepage is resisted and uplift is reduced.
4. **What is the life of a reservoir?** Time taken for sediment to fill the reservoir (storage / annual silt inflow).
5. **How do we reduce reservoir leakage?** Grouting, clay blanket, cut-off wall, and selecting impervious rim.

6. **Purpose of tunnel lining?** Support, water-tightness, smooth flow and protection.
7. **Which attitude of beds is best for tunnelling?** Tunnel axis across (perpendicular to) the strike in strong, dry beds.

Final exam tips

- For Unit V always give: (1) a neat sketch, (2) geological factors in bullet form (rock type - structure - water - weathering - seismicity), (3) one Indian example, (4) remedial measure (grouting / lining).